

Model Poisoning-Resilient Split Federated Learning

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Contents

File 01: Federated Learning Baseline on MNIST (IID vs Extreme Non-IID)

This notebook establishes the first Federated Learning (FL) baseline experiment of the project using the MNIST handwritten digit dataset. The primary objective of this file is to understand how a standard FL system behaves under both ideal IID conditions and extreme non-IID label-skew conditions before introducing Split Federated Learning (SFL), deeper architectures, poisoning attacks, or defence mechanisms. The file focuses entirely on standard FL behaviour and provides the foundational reference point for all subsequent experiments in the project.

The implemented model is a shallow fully connected neural network (TwoNN-style architecture). The network consists of an input flattening stage followed by a hidden fully connected layer with ReLU activation and a final output layer containing 10 output classes corresponding to the MNIST digits. The FL system uses standard Federated Averaging (FedAvg) aggregation, where clients train locally using SGD optimisation before transmitting updated model weights to the server for global aggregation. The experiments use 10 distributed clients and 100 communication rounds.

Two separate training scenarios were investigated. In Stage 1, the training data was distributed using an IID partitioning strategy, where each client receives a balanced random subset of the dataset. In Stage 2, an extreme non-IID configuration was implemented where each client receives data from only a single label class (one-label-per-client). This creates severe client heterogeneity and simulates one of the most difficult distributed learning environments for FL optimisation.

The IID experiment demonstrated stable and efficient convergence, reaching approximately 0.97 test accuracy after 100 communication rounds. In contrast, the extreme non-IID experiment converged much more slowly and stabilised around approximately 0.84–0.85 accuracy. Although the one-label-per-client setting significantly reduced convergence speed and final performance, the FL system still successfully learned a reasonably accurate global classifier despite each client only observing a single digit class locally.

Several important observations were obtained from this experiment. First, standard FL remains surprisingly robust under severe label-skew heterogeneity when applied to relatively simple datasets such as MNIST. Second, extreme non-IID conditions introduce substantial optimisation difficulty and slower global convergence due to client drift and conflicting local objectives. Third, the experiment established a strong baseline reference for future comparison with Split Federated Learning architectures, deeper models, poisoning attacks, and robust defence mechanisms explored later in the project.

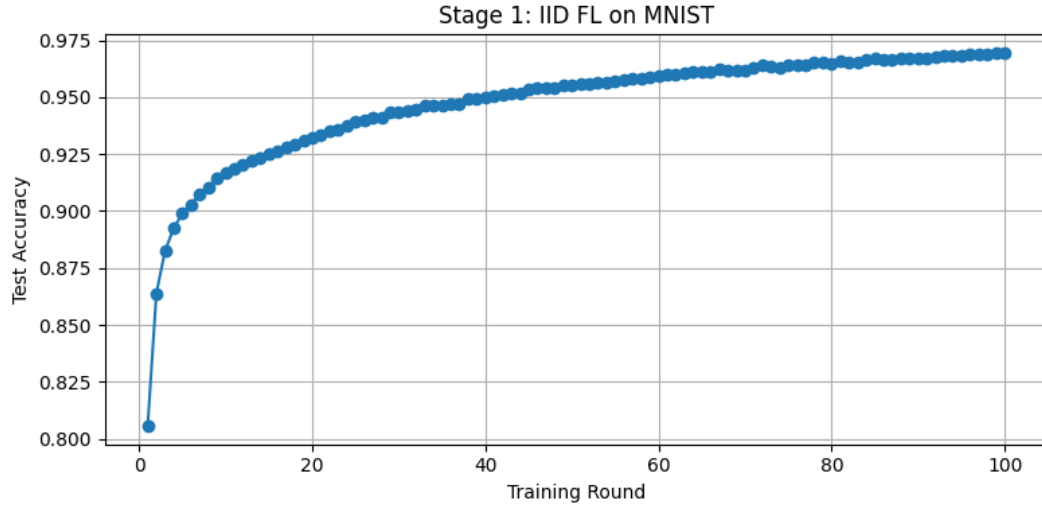


Figure 1: Stage 1: IID Federated Learning performance on MNIST.

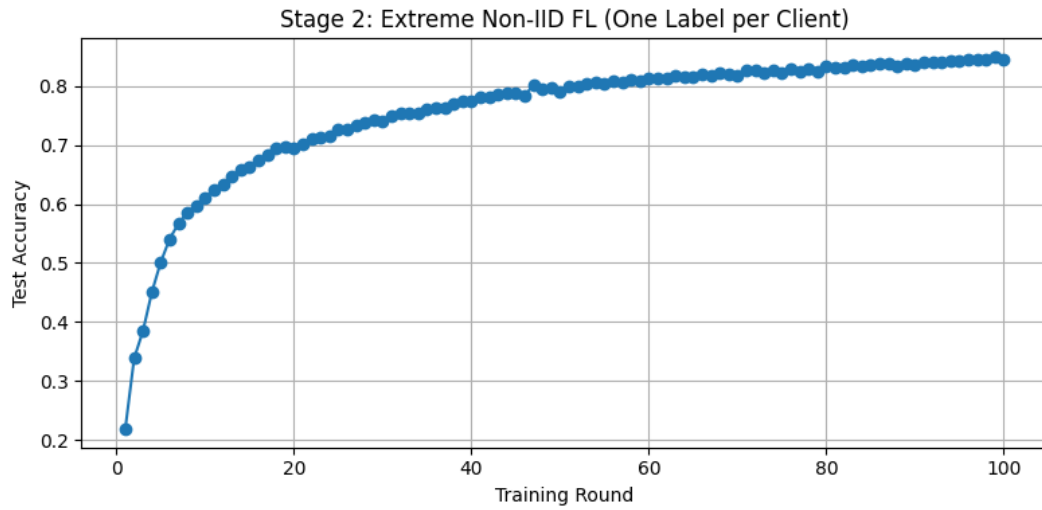


Figure 2: Stage 2: Extreme non-IID Federated Learning performance using one-label-per-client partitioning.

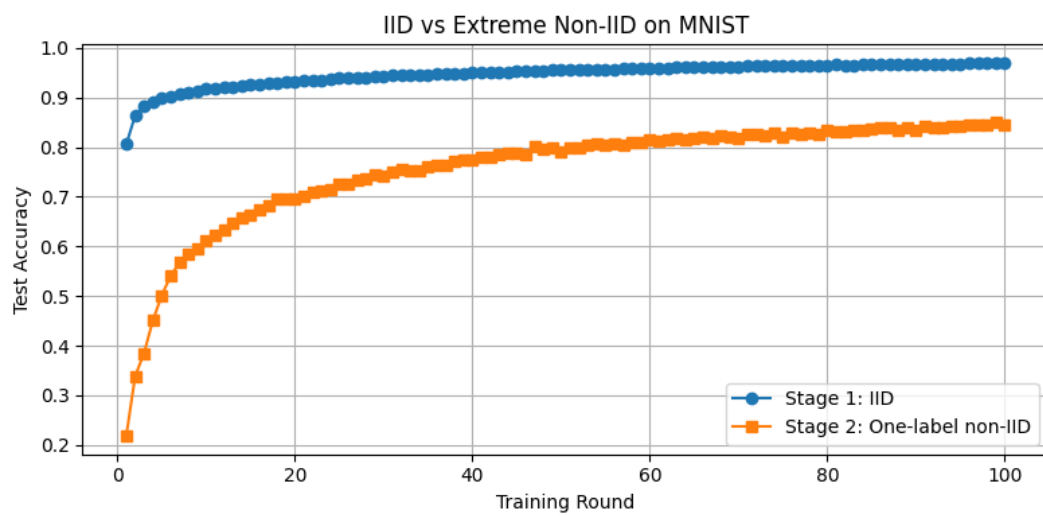


Figure 3: Comparison between IID and extreme non-IID FL performance on MNIST.

File 02: Federated Learning Baseline Across MNIST, Fashion-MNIST, and CIFAR-10

This notebook establishes the initial Federated Learning (FL) baseline for the project. The objective of this file is to investigate how standard FL behaves under both IID and highly non-IID client data distributions before introducing Split Federated Learning (SFL), poisoning attacks, or defence mechanisms. The experiments were conducted on three datasets with increasing complexity: MNIST, Fashion-MNIST (FMNIST), and CIFAR-10. A shallow TwoNN fully connected neural network was used as the baseline model. The architecture consisted of an input flattening layer, a single hidden fully connected layer with 200 neurons and ReLU activation, and a final output layer containing 10 output classes. The model was intentionally kept shallow in order to first understand the effect of data heterogeneity without the added complexity of deeper architectures.

The FL implementation used 10 distributed clients and standard Federated Averaging (FedAvg) aggregation. Each communication round consisted of clients performing local SGD optimisation on their own local datasets before transmitting their updated model weights to the server for aggregation. Multiple data partitioning strategies were evaluated. In the IID configuration, the training data was randomly and evenly distributed across all clients. In the non-IID configurations, each client was restricted to only a subset of labels, specifically 1, 2, 4, or 6 labels per client. This produced increasingly severe label-skew heterogeneity and simulated realistic distributed learning environments where clients may observe only highly specialised local data.

The experiments demonstrated that non-IID data significantly affects FL convergence and final model performance, although the severity of the degradation depends strongly on dataset complexity. MNIST produced the strongest results, with IID training converging close to 0.98 accuracy and the extreme 1-label-per-client setting still achieving approximately 0.80 accuracy after 100 communication rounds. Fashion-MNIST was more difficult, with IID performance converging near 0.89 accuracy and the 1-label-per-client setting converging around 0.70–0.72 accuracy. CIFAR-10 proved substantially more challenging for the shallow fully connected TwoNN architecture. IID accuracy reached only approximately 0.50, while the 1-label-per-client setting converged around 0.30 accuracy. The 2-label-per-client CIFAR-10 configuration also remained highly unstable and close to random classification performance for much of the training process.

The results of this file produced several important observations. First, shallow neural network architectures appear surprisingly robust under extreme non-IID conditions on simpler datasets such as MNIST. Second, increasing dataset complexity significantly amplifies the optimisation difficulty introduced by client heterogeneity. Third, assigning additional labels per client consistently improves convergence stability and final global accuracy. These findings established the baseline behaviour of standard FL under severe non-IID conditions and motivated the next stage of the project, which focused on validating Split Learning equivalence and exploring how architectural partitioning affects convergence, poisoning robustness, and defence behaviour.

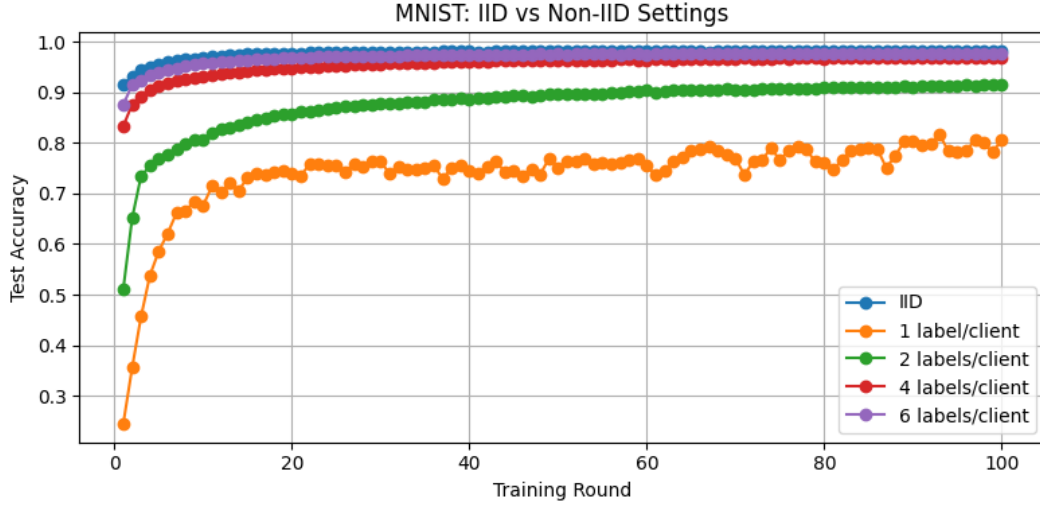


Figure 4: MNIST IID vs Non-IID FL performance using the TwoNN architecture.

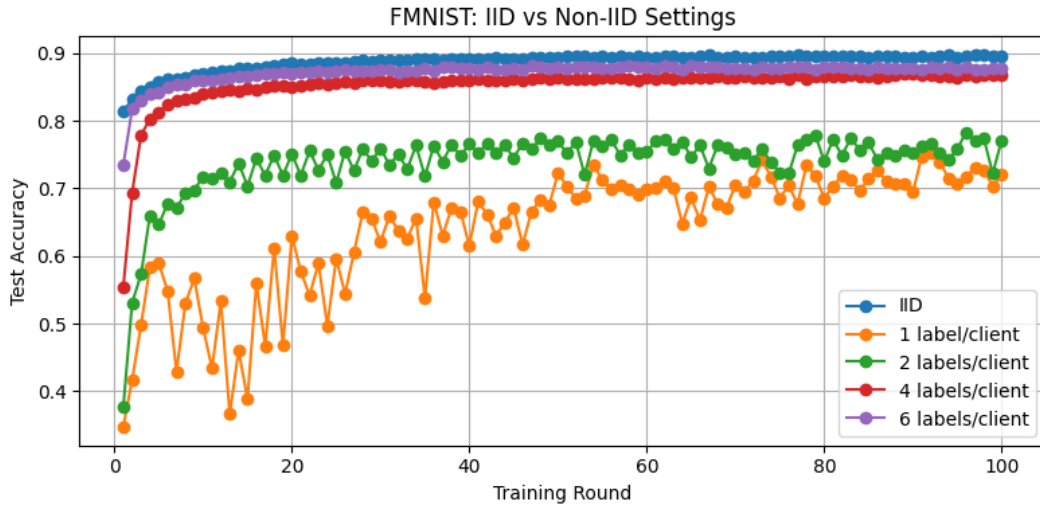


Figure 5: Fashion-MNIST IID vs Non-IID FL performance using the TwoNN architecture.

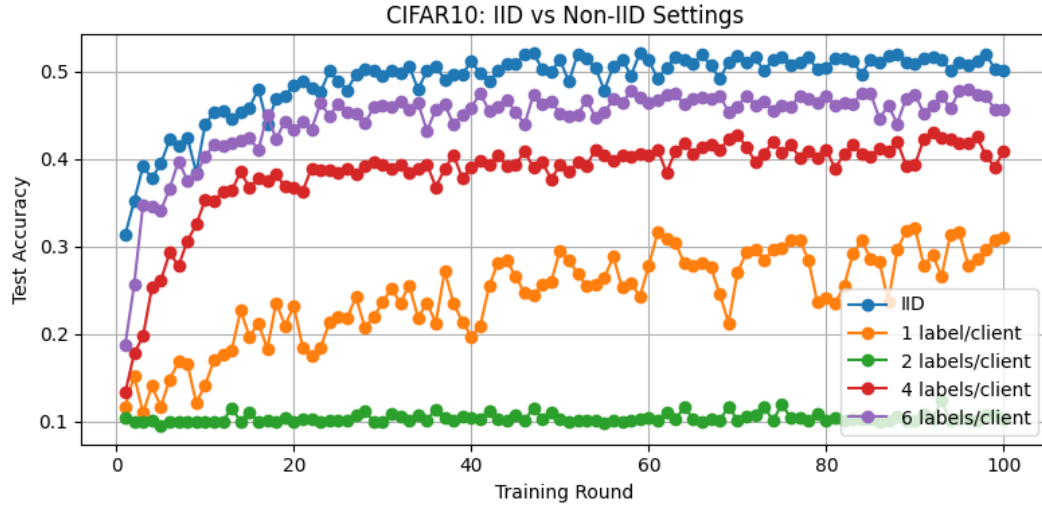


Figure 6: CIFAR-10 IID vs Non-IID FL performance using the TwoNN architecture.

File 03: Split Learning vs Full Model Comparison

This experiment implemented and compared a standard centralized neural network against a Split Learning (SL) version of the same architecture. The objective was to verify whether splitting the model between a client and server changes the learning behaviour or final model performance under controlled IID conditions.

The experiment was performed on the MNIST dataset using a single client. The neural network was divided into:

- Client side: first 2 layers
- Server side: remaining 3 layers

Both the full model and split model used:

- identical initialization,
- identical optimizer settings,
- identical learning rate,
- and the same training data.

The purpose of this setup was to ensure that any observed differences would result only from the split architecture itself.

Results

The results showed that the split model achieved nearly identical behaviour to the centralized full model:

- Test accuracy converged to approximately 97–98%
- Test loss curves almost completely overlapped
- Both models followed nearly identical convergence behaviour throughout training

These results confirm that Split Learning can reproduce centralized learning performance under IID conditions when implemented correctly.

The overlapping accuracy and loss curves indicate that gradient propagation across the client-server boundary was functioning correctly. The experiment demonstrated that partitioning the model between client and server does not inherently reduce learning capability.

This notebook therefore established the correctness of the Split Learning pipeline before extending the work toward Split Federated Learning (SFL), multiple clients, Non-IID distributions, poisoning attacks, and defence mechanisms.

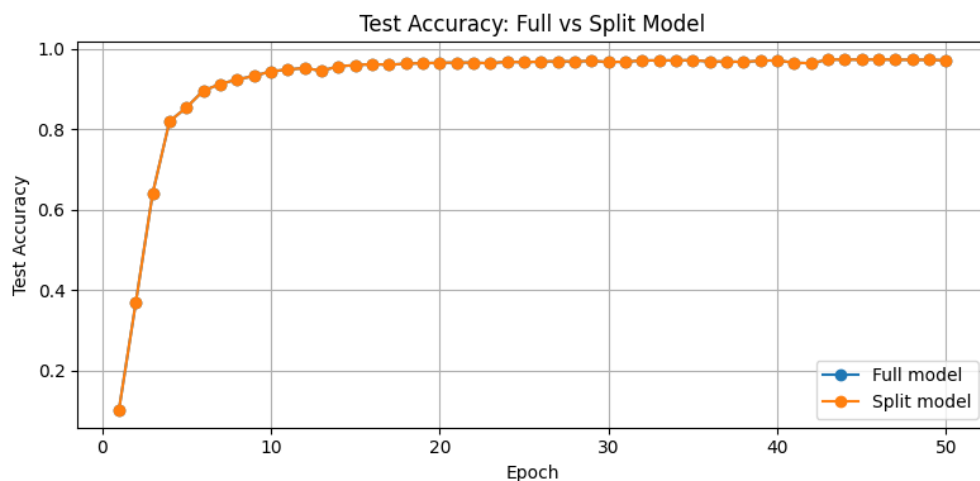


Figure 7: Test Accuracy: Full Model vs Split Model

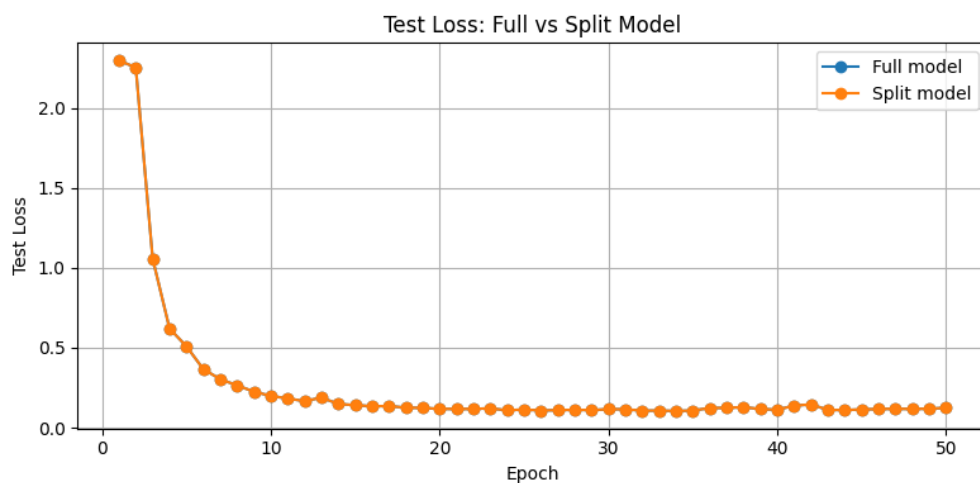


Figure 8: Test Loss: Full Model vs Split Model

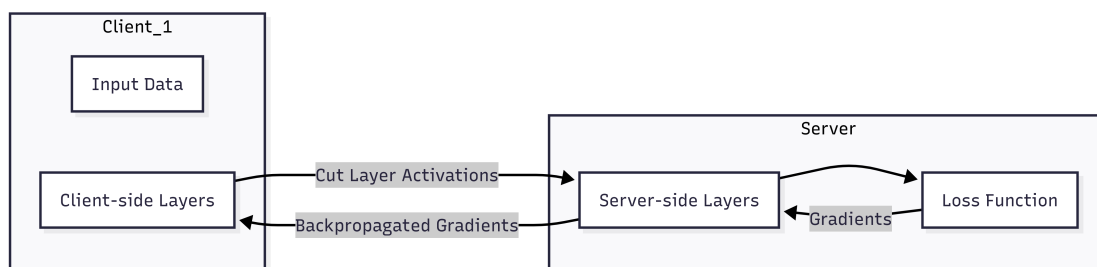


Figure 9: Split Federated Learning architecture.

File 04: Federated Learning vs Split Learning Equivalence Study (3 Clients, MNIST)

This notebook investigates the mathematical equivalence between Federated Learning (FL) and Split Federated Learning (SFL) under controlled deterministic conditions using the MNIST dataset. The purpose of this file is to validate that the implemented SFL framework behaves identically to standard FL when both systems are initialised with identical weights, use identical client datasets, and process data in the exact same order. This experiment forms a critical validation stage in the project because it establishes confidence that any future divergence between FL and SFL is caused by architectural behaviour, attacks, or defence mechanisms rather than implementation inconsistencies.

The experiment uses three distributed clients and a shallow fully connected neural network architecture trained over 50 communication rounds. In the standard FL implementation, each client trains a complete local model and transmits updated model parameters to the server where Federated Averaging (FedAvg) aggregation is performed. In the SFL implementation, the model is partitioned into client-side and server-side components. The client computes the forward pass through the client-side layers and transmits activations to the server, which completes the forward and backward propagation process before returning gradients to the client. Separate aggregation is then performed for the client-side and server-side model partitions before reconstructing a complete global SFL model.

A major feature of this notebook is the strict enforcement of deterministic training conditions. Random seeds, initial model parameters, client dataset allocation, mini-batch ordering, optimiser behaviour, and aggregation ordering are all synchronised between FL and SFL. This ensures that both systems receive mathematically identical optimisation conditions throughout training.

The FL and reconstructed SFL global models are evaluated independently after every communication round using the MNIST test dataset. Test accuracy and test loss are computed using the same evaluation pipeline for both systems. The reconstructed SFL model is formed by combining the aggregated client-side layers and aggregated server-side layers into a single equivalent network representation for direct comparison against the FL global model.

The notebook also computes the global L2 distance between the FL global model parameters and reconstructed SFL global model parameters. The L2 distance is calculated using:

$$L_2 = \|\theta_{FL} - \theta_{SFL}\|_2$$

where:

$$\theta_{FL}$$

represents the FL global model parameters and

$$\theta_{SFL}$$

represents the reconstructed SFL global model parameters. All model parameters are

flattened into vectors before computing the Euclidean norm. If both systems remain mathematically equivalent throughout optimisation, the L2 distance should remain exactly zero.

The results demonstrate complete equivalence between FL and SFL under controlled conditions. The global accuracy curves overlap perfectly throughout all 50 communication rounds, converging to approximately 0.96–0.97 test accuracy. Similarly, the global test loss curves are numerically identical across all rounds and decrease steadily from approximately 2.3 to near 0.12. Most importantly, the global L2 distance remains exactly zero throughout the entire experiment, confirming that the reconstructed SFL global model is mathematically identical to the FL global model at every communication round.

The notebook additionally records timing information for both FL and SFL training. Separate timing measurements are maintained for FL client training, SFL client/server processing, aggregation, and evaluation overhead. These timing metrics establish a baseline computational comparison that becomes important in later poisoning attack and defence experiments where additional anomaly detection logic introduces extra computational cost.

This file establishes one of the most important foundations of the project. By proving exact equivalence between FL and SFL under deterministic conditions, it validates the correctness of the SFL implementation and ensures that future behavioural differences observed under non-IID distributions, poisoning attacks, or defence mechanisms are genuine architectural effects rather than numerical inconsistencies or implementation errors.

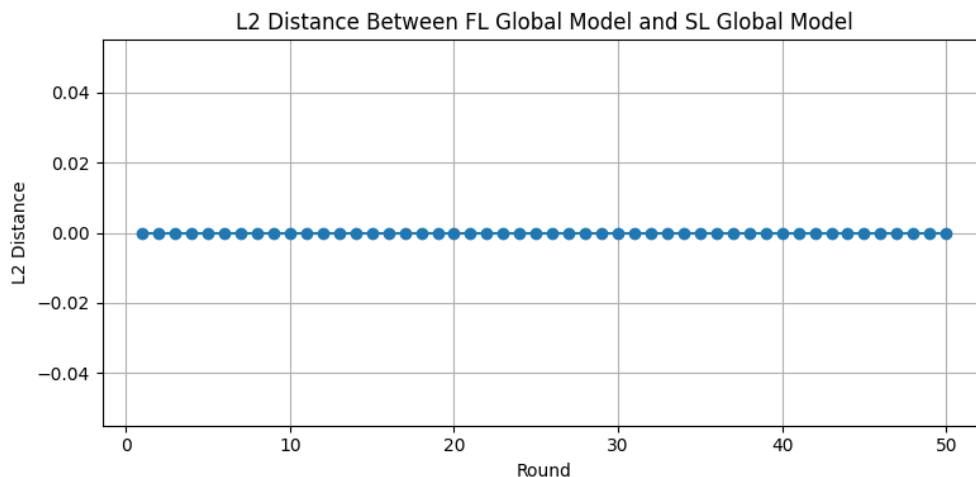


Figure 10: Global L2 distance between the FL global model and reconstructed SFL global model. The L2 distance remains exactly zero throughout all communication rounds, demonstrating mathematical equivalence.

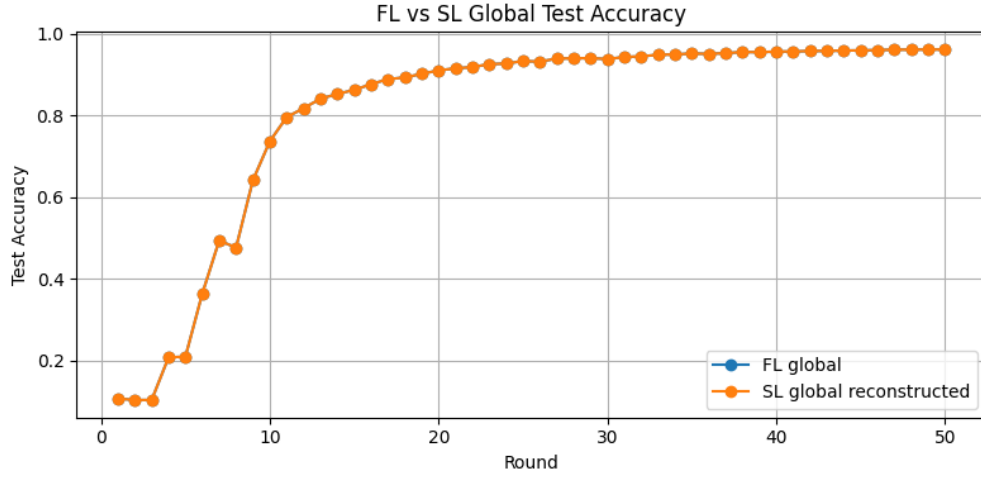


Figure 11: Comparison of FL and reconstructed SFL global test accuracy over 50 communication rounds. Both systems produce identical convergence behaviour.

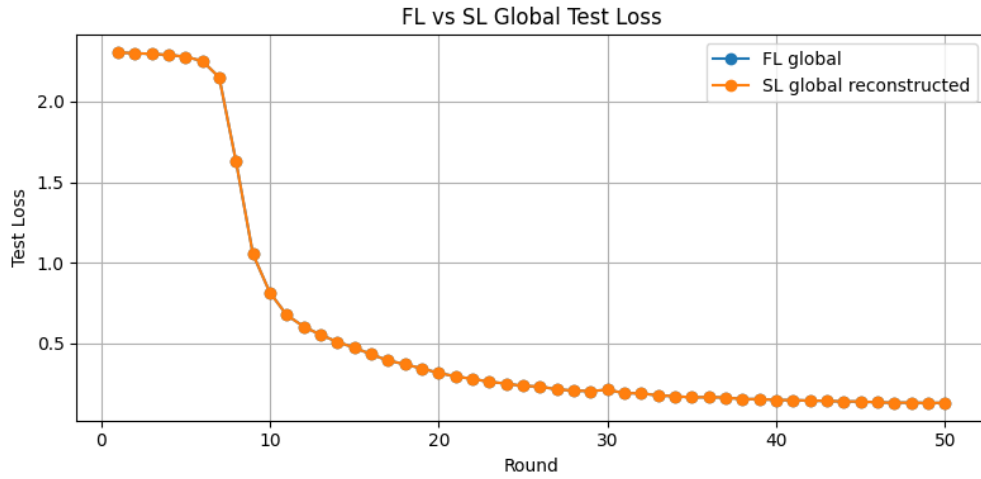


Figure 12: Comparison of FL and reconstructed SFL global test loss over 50 communication rounds. Both systems produce identical optimisation trajectories.

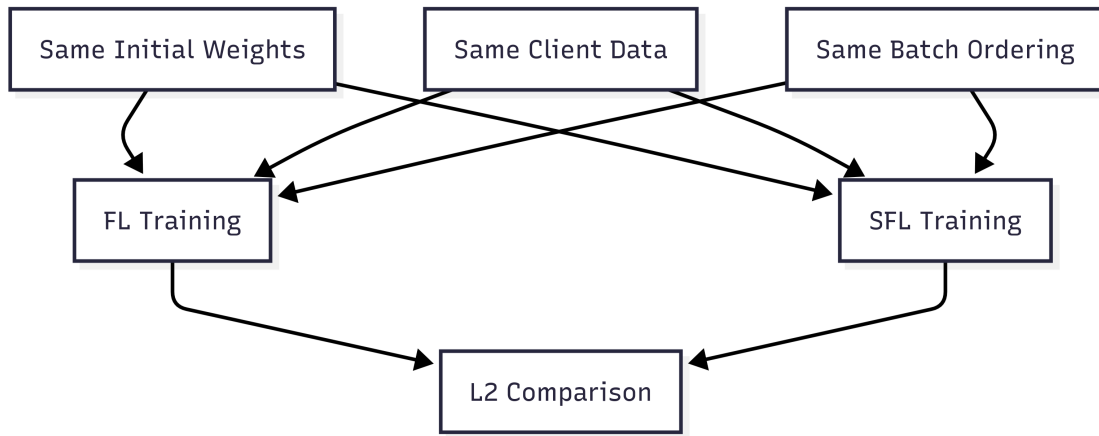


Figure 13: Controlled FL vs SFL equivalence experiment setup.

File 05A: FL vs SFL under Extreme Non-IID (Two-Layer Architecture)

This notebook represents the first major transition from the earlier IID equivalence studies into realistic heterogeneous distributed learning conditions. Files 01 and 02 established the baseline behaviour of Federated Learning (FL) under IID and non-IID data distributions using shallow TwoNN architectures. File 03 then validated that Split Learning (SFL) could reproduce standard neural network training behaviour under IID conditions using a split client/server architecture. File 04 extended this further by proving exact mathematical equivalence between FL and SFL under deterministic training conditions, where both systems produced identical global models with zero L2 distance throughout training. Having established this equivalence foundation, File 05A introduces the first extreme non-IID FL vs SFL comparison using one-label-per-client partitioning without any poisoning attacks or defence mechanisms.

The experiment uses the MNIST handwritten digit dataset with 10 distributed clients. Unlike the earlier IID experiments, each client now receives training data from only a single digit class. This creates an extreme label-skew non-IID environment where clients possess highly specialised and non-overlapping local datasets. Such a configuration significantly increases optimisation difficulty because local client gradients no longer represent the global data distribution.

The implemented neural network architecture is a shallow two-layer fully connected model:

$$784 \rightarrow 256 \rightarrow 10$$

where:

- 784 corresponds to the flattened MNIST input image,
- 256 represents the hidden layer dimension,
- and 10 represents the final classification output classes.

This architecture was intentionally selected because it closely resembles the original TwoNN baseline used in earlier FL experiments and had already demonstrated relatively strong robustness under non-IID conditions.

The notebook simultaneously trains two completely separate systems:

1. Standard Federated Learning (FL)
2. Split Federated Learning (SFL)

The FL system follows conventional Federated Averaging (FedAvg). Each client trains a complete local copy of the neural network using local SGD optimisation before transmitting updated model parameters to the server for weighted global aggregation.

The SFL system uses a split architecture where the model is partitioned between client-side and server-side components. In this file, the architecture is divided such that:

- the client executes the earlier network portion,
- and the server executes the remaining layers.

The client performs forward propagation through the client-side layers and sends intermediate activations to the server. The server then completes the forward pass, computes the loss, performs backpropagation, and returns gradients to the client for local parameter updates.

An important design feature of this notebook is that FL and SFL remain fully independent systems throughout training. The FL global model is trained exclusively using FL aggregation, while the SFL reconstructed global model is trained entirely through separate client-side and server-side aggregation mechanisms. Neither system exchanges gradients, parameters, or updates with the other during optimisation. Their only interaction occurs during post-round evaluation where their final global model states are compared analytically.

To ensure fair comparison between FL and SFL, deterministic training conditions are enforced throughout the notebook. Both systems:

- start from identical initial weights,
- use identical client dataset assignments,
- use identical mini-batch ordering,
- and apply identical optimisation schedules.

The notebook additionally maintains a decaying learning rate schedule during training. The learning rate is progressively reduced over communication rounds in order to stabilise optimisation under the highly heterogeneous one-label-per-client setting.

Mathematical Foundation of FL and SFL Equivalence

The mathematical objective of this notebook is to verify that Federated Learning (FL) and Split Federated Learning (SFL) perform equivalent optimisation when trained under identical deterministic conditions. Although the architectures are partitioned differently, both systems ultimately attempt to minimise the same global objective function:

$$\min_{\theta} F(\theta)$$

where:

$$F(\theta) = \sum_{k=1}^K \frac{n_k}{n} F_k(\theta)$$

and:

- K represents the number of clients,
- n_k represents the number of samples on client k ,

- n is the total number of samples across all clients,
- and $F_k(\theta)$ is the local loss function for client k .

For each client, the local optimisation objective is:

$$F_k(\theta) = \frac{1}{n_k} \sum_{i=1}^{n_k} \ell(f_\theta(x_i), y_i)$$

where:

- x_i is the input sample,
- y_i is the target label,
- $f_\theta(x_i)$ is the neural network prediction,
- and $\ell(\cdot)$ represents the cross-entropy classification loss.

In standard FL, each client independently performs SGD updates on the complete model:

$$\theta_k^{(t+1)} = \theta^{(t)} - \eta \nabla F_k(\theta^{(t)})$$

After local optimisation, the server performs Federated Averaging (FedAvg):

$$\theta^{(t+1)} = \sum_{k=1}^K \frac{n_k}{n} \theta_k^{(t+1)}$$

In SFL, the optimisation process remains mathematically equivalent but computationally partitioned between client-side and server-side layers. The model parameters are divided into:

$$\theta = (\theta_c, \theta_s)$$

where:

- θ_c represents client-side parameters,
- and θ_s represents server-side parameters.

The client computes forward activations:

$$a = f_{\theta_c}(x)$$

which are transmitted to the server. The server then computes:

$$\hat{y} = g_{\theta_s}(a)$$

followed by the classification loss:

$$\ell(\hat{y}, y)$$

During backpropagation, gradients are computed on the server:

$$\frac{\partial \ell}{\partial \theta_s}$$

and activation gradients are returned to the client:

$$\frac{\partial \ell}{\partial a}$$

allowing the client to compute:

$$\frac{\partial \ell}{\partial \theta_c}$$

Although the computation is partitioned, the optimisation trajectory remains mathematically identical to standard FL when:

- initial weights are identical,
- mini-batch ordering is identical,
- client datasets are identical,
- and optimisation hyperparameters remain synchronized.

A major analytical component inherited from File 04 is the comparison between the FL global model and the reconstructed SFL global model using the global L2 distance metric:

$$L_2 = \|\theta_{FL} - \theta_{SFL}\|_2$$

where:

- θ_{FL} represents the FL global model parameters,
- and θ_{SFL} represents the reconstructed SFL global model parameters.

The Euclidean norm is computed by flattening all parameters into vectors:

$$L_2 = \sqrt{\sum_{i=1}^d (\theta_{FL,i} - \theta_{SFL,i})^2}$$

where:

- d represents the total number of parameters in the model.

If both systems remain mathematically equivalent throughout optimisation, the L2 distance should remain exactly zero.

Results and Observations

The results demonstrate that both FL and SFL converge identically despite the highly fragmented client distributions. The global accuracy curves overlap perfectly and converge to approximately 0.78 test accuracy after 100 communication rounds. The global loss curves also overlap exactly and decrease steadily from approximately 2.23 to below 0.95 throughout training.

Several important observations emerge from this experiment. First, shallow neural network architectures remain surprisingly trainable even under extremely heterogeneous one-label-per-client conditions. Second, the mathematical equivalence between FL and SFL established in File 04 continues to hold under severe non-IID optimisation settings. Third, maintaining identical deterministic conditions ensures that any future divergence observed in later poisoning or defence experiments can be confidently attributed to attack or defence behaviour rather than implementation inconsistencies.

This file therefore forms the final clean baseline before introducing poisoning attacks and robust defence mechanisms in subsequent experiments.

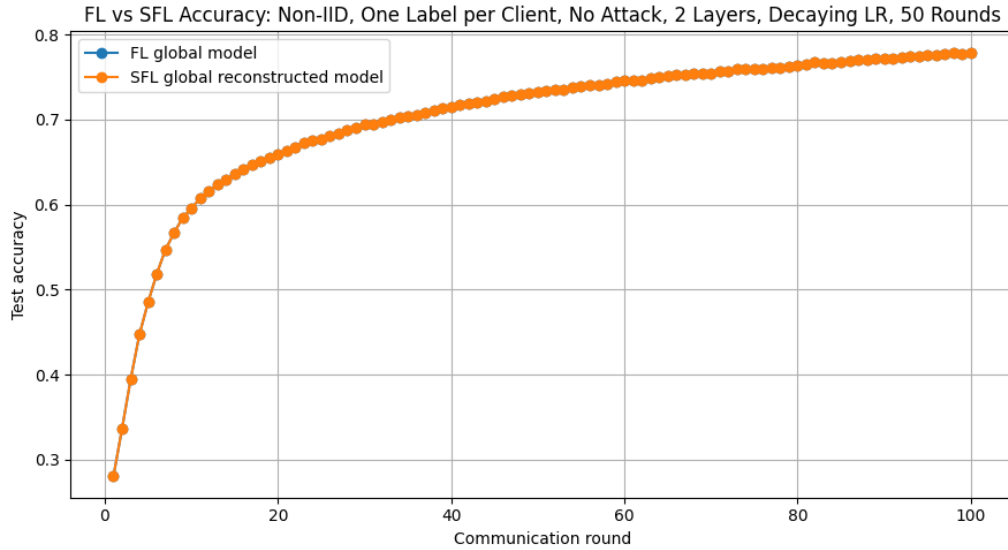


Figure 14: FL vs reconstructed SFL global test accuracy under extreme non-IID one-label-per-client training using the shallow two-layer architecture. Both systems converge identically to approximately 0.78 accuracy after 100 communication rounds.

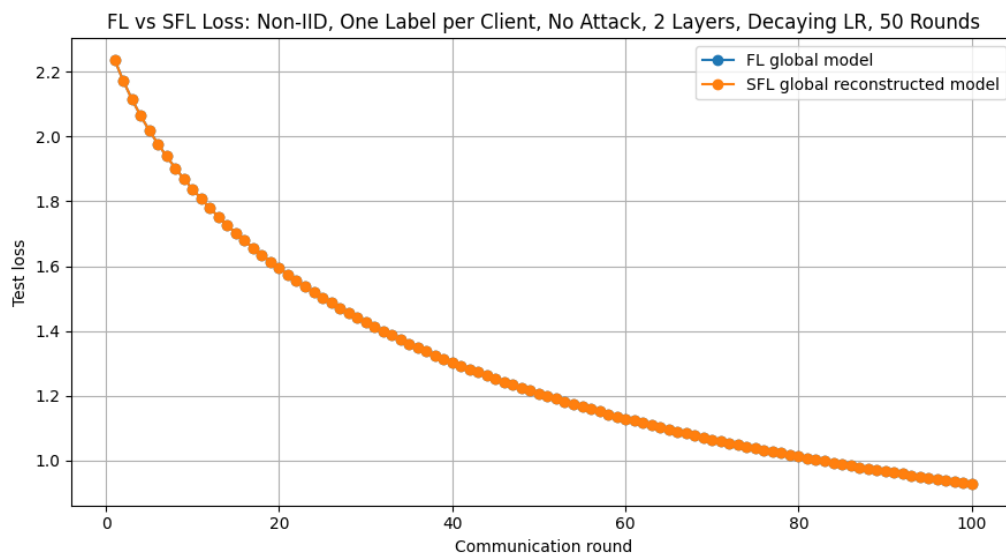


Figure 15: FL vs reconstructed SFL global test loss under extreme non-IID one-label-per-client training. Both systems exhibit identical optimisation behaviour and steadily decreasing loss throughout training.

File 05B: FL vs SFL under Extreme Non-IID (Three-Layer Architecture)

Following the successful equivalence validation achieved in File 05A using the shallow two-layer architecture, this notebook extends the investigation by increasing the model depth to a three-layer neural network architecture. The objective of this file is to study how increasing architectural complexity affects optimisation behaviour under extreme non-IID one-label-per-client conditions while maintaining equivalence between Federated Learning (FL) and Split Federated Learning (SFL).

The motivation for progressing from File 05A to File 05B was based on several observations from the earlier experiments. The shallow two-layer architecture demonstrated strong convergence and stability even under severe client heterogeneity, reaching approximately 0.78 test accuracy after 100 communication rounds. However, the model structure remained relatively simple and provided only a minimal split configuration for SFL. To better represent realistic Split Federated Learning behaviour and prepare for later poisoning attack experiments, a deeper architecture was required where the client-side and server-side model partitions became more meaningful.

The architecture implemented in this notebook is a three-layer fully connected neural network:

$$784 \rightarrow 256 \rightarrow 128 \rightarrow 10$$

where:

- 784 corresponds to the flattened MNIST image input,
- 256 and 128 represent hidden layer dimensions,
- and 10 represents the output classification layer.

Unlike File 05A, which used a simpler split, this notebook partitions the architecture into:

- one client-side layer,
- and two server-side layers.

This creates a more representative Split Federated Learning setup where the server performs a larger portion of the computation and parameter optimisation process.

The experiment continues to use the MNIST dataset under the extreme non-IID one-label-per-client partitioning strategy. Ten distributed clients are simulated, where each client receives samples belonging to only a single digit class. As in File 05A, both FL and SFL are trained completely independently under identical deterministic conditions. Both systems:

- start from identical initial weights,
- use identical client dataset assignments,
- use identical mini-batch ordering,

- and apply identical optimisation schedules.

This notebook also introduces several optimisation refinements compared to File 05A. A decaying learning rate schedule is maintained throughout training in order to improve stability under the deeper architecture. Separate client-side and server-side aggregation mechanisms are also explicitly maintained for the SFL implementation, further reinforcing the independence between FL and SFL training dynamics.

As established in File 04 and File 05A, the FL global model and reconstructed SFL global model remain mathematically equivalent throughout training. The reconstructed SFL global model is formed by combining the aggregated client-side layers and aggregated server-side layers into a complete network representation. The global L2 distance between FL and reconstructed SFL is continuously monitored to verify numerical equivalence between the two optimisation systems.

The results demonstrate that both FL and SFL again converge identically throughout training, confirming that the mathematical equivalence between FL and SFL continues to hold even after increasing architectural depth. However, compared to File 05A, several important behavioural changes are observed.

The global accuracy converges to approximately 0.72–0.73 after 100 communication rounds, which is noticeably lower than the approximately 0.78 accuracy achieved by the shallower two-layer architecture in File 05A. Additionally, the optimisation trajectory becomes more oscillatory and less stable during intermediate communication rounds. The loss curve decreases steadily throughout training but remains substantially higher than the final loss achieved by the shallower architecture.

These results suggest that increasing model depth significantly increases optimisation difficulty under extreme one-label-per-client non-IID conditions. Because each client only observes a single label class, deeper architectures appear more sensitive to client drift and conflicting local optimisation directions. The additional representational capacity introduced by the deeper model does not compensate for the severe heterogeneity of the client distributions.

Despite the reduction in final accuracy, this architecture was selected as the primary baseline for later poisoning attack experiments. The reason for this decision is that the three-layer structure provides a much more meaningful Split Federated Learning partition, where:

- the client controls only the earlier feature extraction layer,
- while the server controls the deeper representation and classification layers.

This creates a more realistic attack surface for later poisoning attack studies and defence mechanism development.

Overall, File 05B establishes an important transition point in the project. The notebook demonstrates that:

- FL and SFL equivalence continues to hold under deeper architectures,
- increasing model depth amplifies optimisation difficulty under extreme non-IID conditions,

- and shallow architectures may actually be more robust than deeper architectures when clients possess highly fragmented local data distributions.

These observations directly motivated the later poisoning attack experiments and defence mechanism investigations developed in subsequent files.

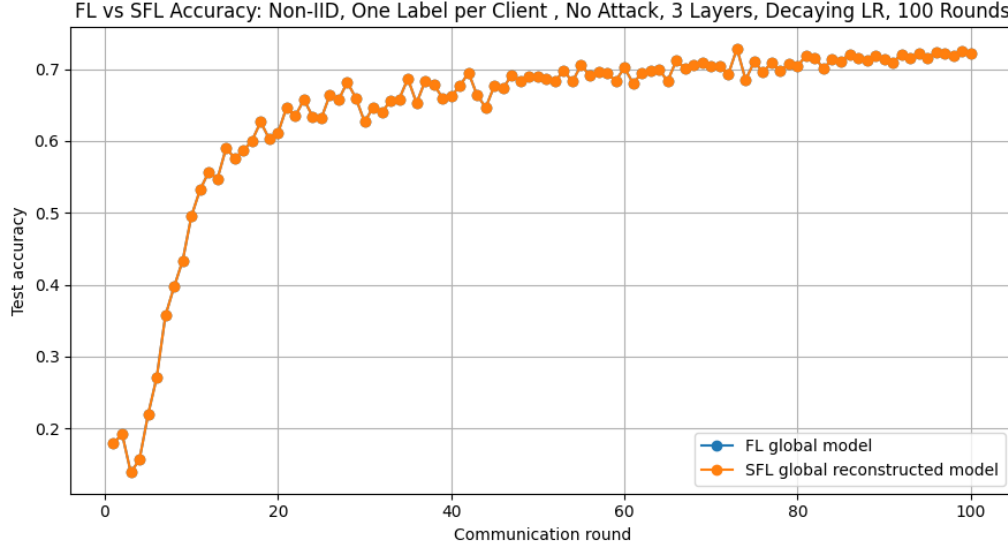


Figure 16: FL vs reconstructed SFL global test accuracy under extreme non-IID one-label-per-client training using the deeper three-layer architecture. Both systems converge identically but achieve lower final accuracy compared to the shallower two-layer model from File 05A.

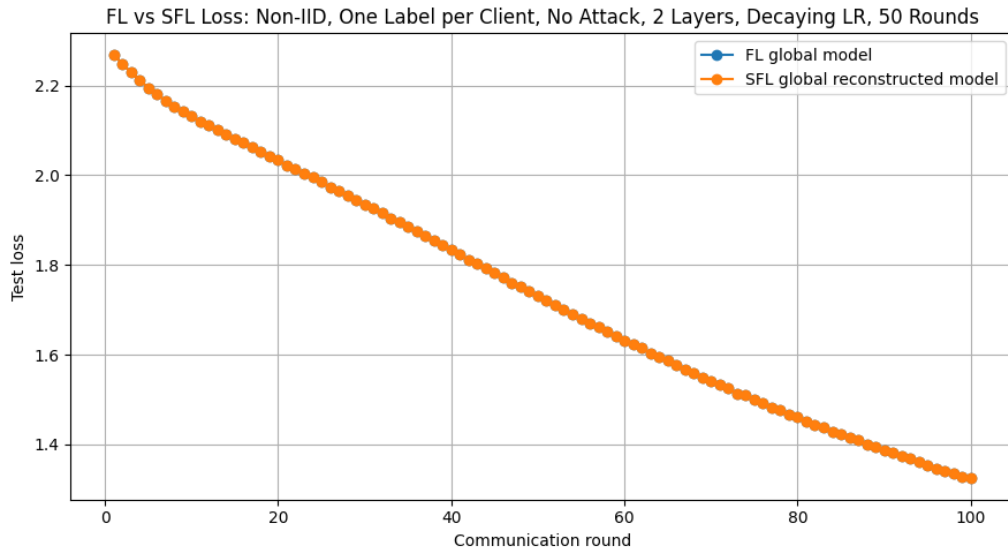


Figure 17: FL vs reconstructed SFL global test loss under extreme non-IID one-label-per-client training using the three-layer architecture. The optimisation trajectory remains stable but converges more slowly compared to the shallower architecture.

File 05C: FL vs SFL under Extreme Non-IID using Fashion-MNIST

After establishing in File 05B that deeper three-layer architectures remain mathematically equivalent between Federated Learning (FL) and Split Federated Learning (SFL) under extreme one-label-per-client conditions, the next logical progression was to investigate whether the same behaviour would persist when the underlying dataset became significantly more complex. File 05C therefore replaces the MNIST handwritten digit dataset with the Fashion-MNIST (FMNIST) dataset while maintaining the same overall FL and SFL architecture and experimental design.

The motivation for this transition was based on an important limitation observed in earlier experiments. MNIST is a relatively simple and highly separable dataset where many classes possess distinct visual structures. Although the one-label-per-client partitioning created severe heterogeneity, the classification task itself remained comparatively easy. Fashion-MNIST introduces substantially greater class overlap and visual similarity between categories such as shirts, pullovers, coats, and t-shirts. As a result, local client optimisation under one-label-per-client conditions becomes much more difficult because clients train on highly specialised but visually ambiguous distributions.

The experiment again uses ten distributed clients under an extreme non-IID setting where each client receives training samples belonging to only a single class. However, unlike the earlier MNIST experiments, the local distributions now contain significantly more feature overlap between classes that are not visible to individual clients during training.

The neural network architecture remains identical to File 05B:

$$784 \rightarrow 256 \rightarrow 128 \rightarrow 10$$

with:

- a single client-side layer,
- and two server-side layers.

Maintaining the same architecture was important because it allowed the experiment to isolate the effect of dataset complexity while keeping the FL and SFL optimisation structures unchanged.

As in the previous files, the notebook trains two completely independent systems:

1. a standard Federated Learning (FL) system,
2. and a Split Federated Learning (SFL) system.

Both systems:

- begin from identical initial weights,
- use identical client dataset partitions,
- use identical mini-batch ordering,
- and apply the same learning rate decay schedule.

The reconstructed SFL global model is again formed by combining the aggregated client-side layers and aggregated server-side layers into a single complete neural network representation. The global L2 distance between the FL global model and reconstructed SFL model continues to be monitored in the same manner as earlier files in order to verify that both systems remain mathematically equivalent throughout optimisation.

Since the deterministic equivalence framework established in Files 04, 05A, and 05B remains unchanged, the L2 calculation methodology is identical and therefore not repeated in detail in this section. The key focus of this experiment is instead the behavioural impact of increased dataset complexity under extreme heterogeneity.

The results show a significant reduction in overall optimisation performance compared to the earlier MNIST experiments. The final test accuracy converges to approximately 0.64–0.65 after 100 communication rounds, compared to approximately 0.72–0.73 achieved in File 05B using MNIST. The training process also becomes substantially more unstable, with large oscillations in both accuracy and loss throughout optimisation.

Several important observations emerge from these results.

First, the equivalence between FL and SFL continues to hold even under substantially more difficult optimisation conditions. The FL and reconstructed SFL accuracy curves overlap throughout training, confirming that the split architecture itself does not introduce optimisation inconsistency.

Second, the combination of:

- extreme non-IID distributions,
- deeper architectures,
- and increased dataset complexity

creates a highly unstable optimisation environment. Because each client observes only one Fashion-MNIST class, local gradients become strongly biased toward narrow local feature representations that generalise poorly to the unseen global distribution.

Third, the experiment demonstrates that deeper architectures under extreme heterogeneity become increasingly sensitive to dataset complexity. Whereas the MNIST experiments remained relatively stable, the Fashion-MNIST experiments exhibit persistent oscillatory behaviour throughout training, particularly during the earlier communication rounds.

This file therefore establishes an important insight for the later poisoning attack studies. It demonstrates that under realistic heterogeneous distributed learning conditions, optimisation instability can already emerge naturally even without adversarial behaviour. This makes robust poisoning attack detection significantly more challenging because the system must distinguish between:

- naturally unstable but legitimate client updates,
- and intentionally malicious poisoned updates.

The observations from this experiment directly motivated the later decision to return to the MNIST dataset in subsequent files before introducing poisoning attacks and defence mechanisms. Although Fashion-MNIST provided a more realistic optimisation challenge, the

natural instability introduced by the dataset made it difficult to isolate the specific effects of adversarial poisoning behaviour.

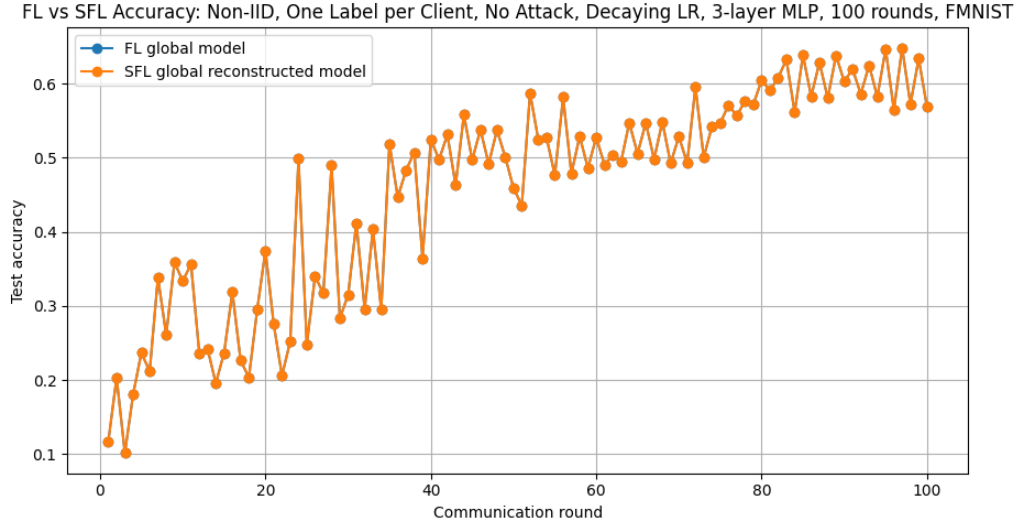


Figure 18: FL vs reconstructed SFL global test accuracy under extreme non-IID one-label-per-client training using Fashion-MNIST and the three-layer architecture. Both systems remain equivalent but exhibit significantly greater instability compared to the earlier MNIST experiments.

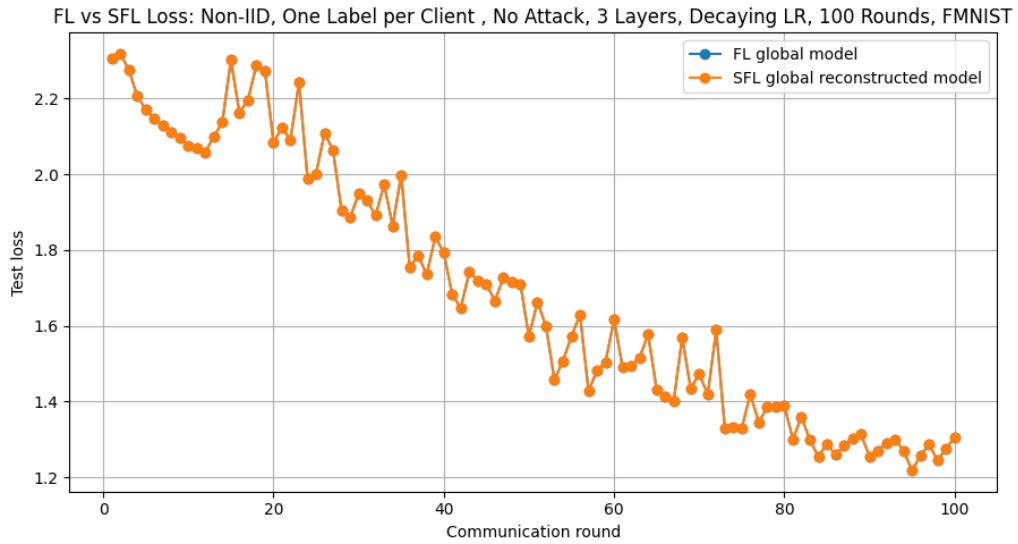


Figure 19: FL vs reconstructed SFL global test loss under extreme non-IID one-label-per-client Fashion-MNIST training. The optimisation trajectory remains highly oscillatory throughout training due to increased dataset complexity and client heterogeneity.

File 05D: FL vs SFL under Extreme Non-IID using CIFAR-10

After investigating the impact of increased dataset complexity using Fashion-MNIST in File 05C, the next stage of the project further increased the learning difficulty by transitioning to the CIFAR-10 dataset. File 05D therefore represents the most challenging clean optimisation experiment conducted prior to introducing poisoning attacks and defence mechanisms.

The primary motivation for progressing from File 05C to File 05D was to study how FL and SFL behave under:

- extreme non-IID client distributions,
- deeper architectures,
- and significantly more complex visual feature spaces.

Whereas Fashion-MNIST still contains grayscale images with relatively structured object shapes, CIFAR-10 introduces:

- coloured RGB images,
- greater intra-class variation,
- lower spatial resolution,
- and substantially more difficult feature extraction requirements.

This creates a much more realistic and difficult distributed optimisation environment.

The dataset used in this notebook is CIFAR-10, consisting of 10 object categories such as:

- airplanes,
- automobiles,
- birds,
- cats,
- dogs,
- ships,
- and trucks.

As in the previous experiments, the data distribution follows an extreme one-label-per-client partitioning strategy. Each distributed client receives training samples belonging to only a single CIFAR-10 class. Under this configuration, clients learn highly specialised local representations with minimal exposure to the global data distribution.

The neural network architecture remains intentionally identical to File 05B and File 05C:

$$3072 \rightarrow 256 \rightarrow 128 \rightarrow 10$$

where:

- 3072 corresponds to the flattened CIFAR-10 RGB image input,
- since CIFAR-10 images contain:

$$32 \times 32 \times 3 = 3072$$

input features,

- 256 and 128 represent hidden layer dimensions,
- and 10 corresponds to the output classification layer.

The Split Federated Learning (SFL) partition also remains unchanged:

- one layer resides on the client side,
- while two layers remain on the server side.

Maintaining the same architecture and split configuration was important because it isolated the effect of increasing dataset complexity while preserving the underlying optimisation structure established in previous files.

The training framework again maintains two completely independent optimisation systems:

1. standard Federated Learning (FL),
2. and Split Federated Learning (SFL).

Both systems:

- begin from identical initial weights,
- use identical client partitions,
- use identical mini-batch ordering,
- and apply the same decaying learning rate schedule.

The reconstructed SFL global model is again formed by combining:

- the aggregated client-side parameters,
- and the aggregated server-side parameters.

The global L2 distance between the FL global model and reconstructed SFL model continues to be monitored throughout training using the same methodology introduced in File 04 and maintained in Files 05A–05C. Since the deterministic equivalence framework remains unchanged, the L2 calculation procedure itself is not modified in this experiment.

The results show a dramatic degradation in optimisation performance compared to all previous datasets. The final global test accuracy converges to only approximately 0.25 after 100 communication rounds, while the loss remains extremely high throughout training.

Several important observations emerge from this experiment.

First, the equivalence between FL and SFL still holds despite the severe optimisation difficulty. The FL and reconstructed SFL accuracy curves continue to overlap, confirming that the split computation structure itself is not responsible for the optimisation instability.

Second, the experiment demonstrates that extreme one-label-per-client heterogeneity becomes increasingly destructive as dataset complexity increases. Unlike MNIST, where local client gradients still contained reasonably separable feature information, CIFAR-10 requires substantially richer feature representations and stronger cross-client knowledge sharing. Under one-label-per-client partitioning, clients are unable to construct meaningful global representations because each client observes only a narrow subset of the feature space.

Third, the three-layer fully connected architecture becomes insufficient for learning CIFAR-10 effectively under such heterogeneous conditions. Unlike convolutional neural networks (CNNs), which can exploit spatial image structure, the fully connected multilayer perceptron (MLP) used in this notebook struggles to extract robust visual features from CIFAR-10 images.

The optimisation trajectory also becomes highly unstable throughout training. Large oscillations appear in both the accuracy and loss curves, and convergence remains slow even after 100 communication rounds. These behaviours indicate that:

- local client gradients are highly conflicting,
- client drift becomes severe,
- and global aggregation struggles to produce coherent optimisation directions.

An important conclusion from this experiment was that combining:

- extreme non-IID distributions,
- deeper architectures,
- fully connected MLPs,
- and highly complex datasets

creates an optimisation environment that is already naturally unstable even without adversarial attacks.

This observation became highly significant for the later poisoning attack experiments. If the clean baseline itself exhibits substantial instability, then distinguishing malicious poisoned behaviour from naturally unstable optimisation becomes extremely difficult. As a result, after completing this experiment, the later poisoning attack and defence studies returned to the MNIST dataset in order to isolate adversarial behaviour more clearly while still maintaining sufficiently challenging non-IID conditions.

Therefore, File 05D represents the final clean optimisation stress test before introducing poisoning attacks and defence mechanisms in the subsequent files.

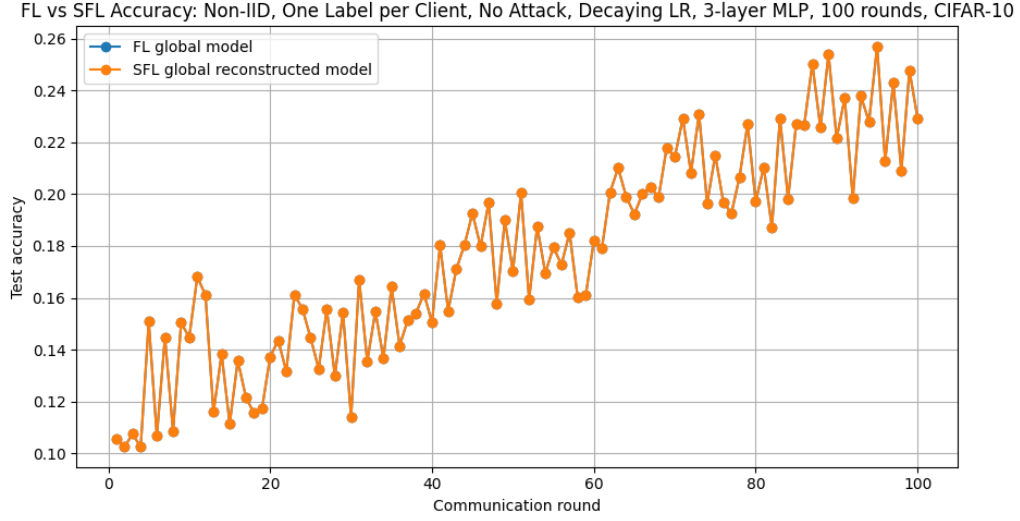


Figure 20: FL vs reconstructed SFL global test accuracy under extreme non-IID one-label-per-client training using CIFAR-10 and the three-layer architecture. Both systems remain equivalent but converge to substantially lower accuracy due to the increased dataset complexity.

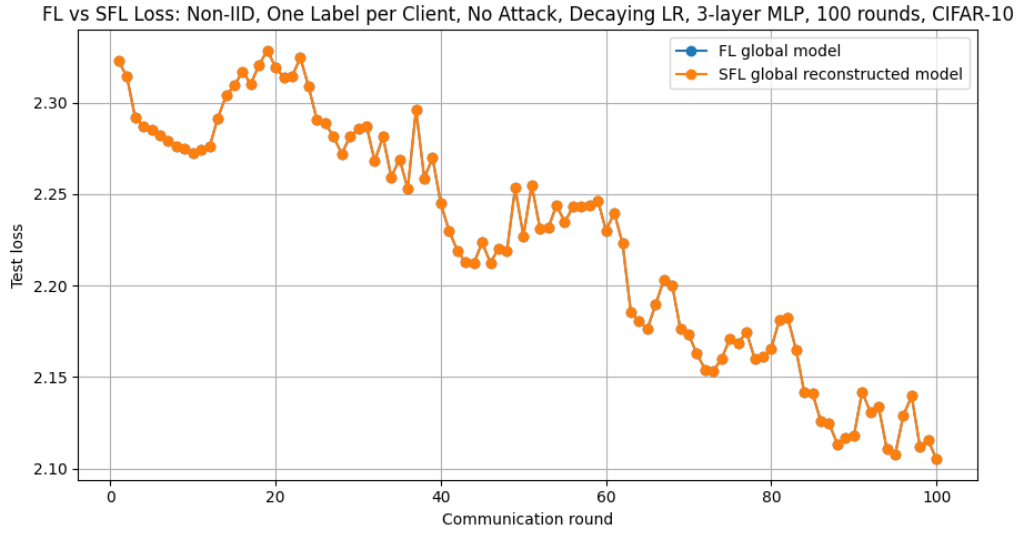


Figure 21: FL vs reconstructed SFL global test loss under extreme non-IID one-label-per-client CIFAR-10 training. The optimisation trajectory remains highly unstable with persistently large loss values throughout training.

File 05E: FL vs SFL under Extreme Non-IID using a Four-Layer Architecture

After completing the progressively more difficult optimisation studies in Files 05A through 05D, the next stage of the project investigated how further increasing architectural depth affects Federated Learning (FL) and Split Federated Learning (SFL) under extreme non-IID one-label-per-client conditions. File 05E therefore introduces the first four-layer multilayer perceptron (MLP) architecture used throughout the project.

The progression to this experiment followed several important observations from the earlier files.

Files 05A and 05B demonstrated that:

- shallow architectures trained relatively well under extreme heterogeneity,
- while deeper architectures became increasingly unstable.

Files 05C and 05D further showed that increasing dataset complexity amplified this instability significantly, especially under:

- one-label-per-client distributions,
- deeper architectures,
- and fully connected networks.

However, although deeper architectures reduced optimisation stability, they also created more meaningful Split Federated Learning partitions where the separation between client-side and server-side computation became increasingly realistic. This was particularly important for the later poisoning attack experiments because:

- attacks in SFL can only directly manipulate the client-side portion of the model,
- while the server-side layers remain protected from direct client access.

The main objective of File 05E was therefore to investigate whether a deeper four-layer architecture could still remain trainable under extreme non-IID conditions while creating a more representative SFL split configuration.

The experiment returns to the MNIST dataset after the severe instability observed in the Fashion-MNIST and CIFAR-10 experiments. Returning to MNIST was a deliberate decision because the later poisoning attack and defence studies required a dataset where:

- the clean optimisation baseline remained sufficiently stable,
- while still preserving challenging non-IID behaviour.

The experiment again uses:

- 10 distributed clients,
- extreme one-label-per-client partitioning,

- and deterministic training conditions.

The architecture implemented in this notebook is a deeper four-layer MLP:

$$784 \rightarrow 256 \rightarrow 128 \rightarrow 10 \rightarrow 10$$

where:

- 784 corresponds to the flattened MNIST input,
- 256 and 128 are hidden layer dimensions,
- and the final two layers both use dimension 10.

An important architectural modification introduced in this notebook is that the final two layers were intentionally reduced to size 10. Earlier experiments showed that increasing model depth under extreme non-IID conditions quickly caused optimisation collapse. The reduction of the final layer dimensions was therefore introduced as a stabilisation mechanism to limit parameter growth and reduce optimisation instability.

The Split Federated Learning partition also changes compared to previous experiments. Unlike File 05B where:

- one layer resided on the client,
- and two layers resided on the server,

File 05E uses:

- two client-side layers,
- and two server-side layers.

This creates a more balanced split architecture where the client performs a larger portion of the feature extraction process before transmitting activations to the server.

As in all previous clean-equivalence experiments, the notebook simultaneously trains:

1. a standard Federated Learning (FL) system,
2. and a Split Federated Learning (SFL) system.

Both systems remain completely independent throughout optimisation and only interact during post-round evaluation and analytical comparison.

The deterministic equivalence framework also remains unchanged:

- identical initial weights,
- identical client partitions,
- identical mini-batch ordering,
- and identical optimisation schedules.

The reconstructed SFL global model is again formed by combining:

- aggregated client-side parameters,
- and aggregated server-side parameters.

The global L2 distance between the FL global model and reconstructed SFL model continues to be calculated using the same Euclidean parameter distance metric established in the earlier equivalence experiments. Since the mathematical procedure remains unchanged, the detailed formulation is not repeated in this section.

The results reveal several important observations.

First, FL and SFL continue to remain mathematically equivalent throughout optimisation. The global accuracy and loss curves overlap almost perfectly throughout training, confirming that increasing architectural depth does not break the equivalence relationship between FL and SFL.

Second, increasing architectural depth further reduces optimisation efficiency under extreme one-label-per-client conditions. Compared to File 05B, which achieved approximately:

0.72 accuracy after 100 rounds,

the four-layer architecture in this notebook achieves approximately:

0.67 – 0.68 accuracy after 200 rounds.

This means that:

- doubling the communication rounds,
- while increasing architectural depth,

still does not recover the optimisation performance achieved by the shallower three-layer architecture.

Third, the optimisation trajectory becomes substantially slower. Although the loss decreases steadily throughout training, convergence occurs gradually over a much longer communication horizon.

In general, these observations are largely consistent with expectations under extreme heterogeneous learning conditions. Under one-label-per-client distributions:

- deeper architectures introduce more parameters,
- require richer shared feature representations,
- and become more sensitive to client drift.

Because each client only observes a single class, the local optimisation process cannot effectively coordinate the deeper feature hierarchies required by larger networks.

However, despite the reduced optimisation efficiency, this experiment remained highly important for the overall project. It demonstrated that:

- deeper SFL partitions remain trainable,

- mathematical equivalence between FL and SFL continues to hold,
- and deeper split architectures create a more realistic attack surface for later poisoning attack studies.

Nevertheless, the experiment also showed that increasing depth beyond the three-layer architecture produced diminishing optimisation returns under extreme non-IID conditions. As a result, after completing this notebook, the project returned to the three-layer MNIST architecture for the subsequent poisoning attack and defence experiments. The three-layer model provided the best compromise between:

- optimisation stability,
- representational complexity,
- and meaningful client/server partitioning.

Thus, File 05E serves as the final clean architectural scaling experiment before transitioning into adversarial poisoning attack investigations.

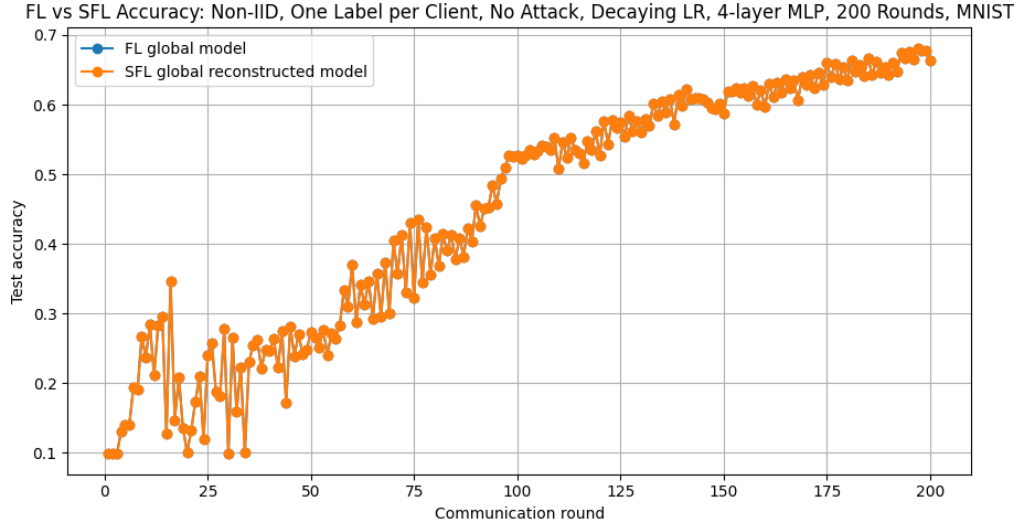


Figure 22: FL vs reconstructed SFL global test accuracy under extreme non-IID one-label-per-client training using the deeper four-layer architecture. Although optimisation remains stable, convergence becomes significantly slower compared to the shallower architectures used in earlier files.

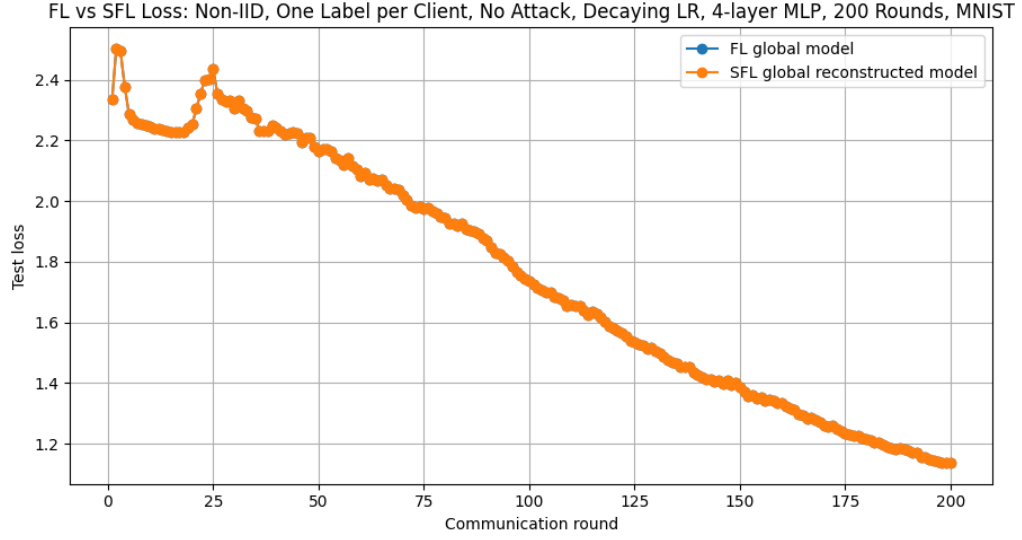


Figure 23: FL vs reconstructed SFL global test loss under extreme non-IID one-label-per-client training using the four-layer architecture. The loss decreases steadily throughout training but converges more slowly than the earlier three-layer experiments.

File 05F: SGA Poisoning Attack on FL and SFL

After completing the clean baseline experiments in Files 05A–05E, File 05F introduces the first poisoning attack experiment. Based on the previous results, the three-layer MNIST model was selected because it provided the best balance between trainability and a meaningful SFL split. The experiment uses MNIST under the extreme one-label-per-client non-IID setting with 10 clients. The architecture is:

$$784 \rightarrow 256 \rightarrow 128 \rightarrow 10$$

For SFL, the model is split into:

$$\text{Client side: } 784 \rightarrow 256$$

$$\text{Server side: } 256 \rightarrow 128 \rightarrow 10$$

The malicious client is client 0. In FL, the attacker can poison the full model update. In SFL, the attacker can only poison the client-side layer, because the server-side layers are not directly controlled by the client. This makes the SFL attack weaker, but also more difficult to detect because the server only observes partial behaviour.

Attack Implementation

The poisoning attack used in this file is a Scale Gradient Attack (SGA). In FL, the malicious client modifies the full model update:

$$\Delta\theta_{FL}^{attack} = -\alpha_{FL}\Delta\theta_{FL}$$

where:

$$\alpha_{FL} = 0.01$$

Although the FL attack scale appears small, it affects the full model update, including all layers. This makes the attack very destructive.

In SFL, the attack is applied only to the client-side update:

$$\Delta\theta_c^{attack} = -\alpha_{SFL}\Delta\theta_c$$

where:

$$\alpha_{SFL} = 20.0$$

The larger SFL attack scale was needed because the attacker only controls the first layer. The server-side layers remain outside the attacker’s direct control.

The SFL client-side aggregation is still similar to a small FL process over the client-side model:

$$\theta_c^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_{c,k}^{t+1}$$

The server-side parameters are aggregated separately:

$$\theta_s^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_{s,k}^{t+1}$$

The reconstructed SFL model is then formed by combining:

$$\theta_{SFL}^{t+1} = (\theta_c^{t+1}, \theta_s^{t+1})$$

Diagnostic Parameters

The generated results table includes several useful diagnostic fields. The field `attack_scale_fl` records the scale used for the FL attack, while `attack_scale_sfl` records the scale used for the SFL attack. The field `grad_clipping_enabled` indicates whether gradients were clipped to prevent numerical explosion. In this file, gradient clipping was disabled:

`grad_clipping_enabled = False`

The field `max_grad_norm` would define the maximum allowed gradient norm if clipping were enabled. Since clipping was disabled, it is recorded as:

`max_grad_norm = None`

The metric `fl_minus_sfl_accuracy` is calculated as:

$$A_{diff} = A_{FL} - A_{SFL}$$

A negative value means SFL is outperforming FL. This becomes important later in training, when FL collapses but SFL continues learning.

The global L2 distance is calculated as:

$$L_2 = \|\theta_{FL} - \theta_{SFL}\|_2$$

Unlike the earlier clean equivalence experiments, the L2 distance is now expected to become non-zero because FL and SFL are no longer receiving equivalent attacks. FL is attacked across the full model, while SFL is attacked only on the client-side layer. Therefore, their optimisation paths naturally diverge. Later in training, FL becomes numerically unstable and some L2 values become NaN, which reflects the collapse of the FL model rather than an equivalence failure in the clean setup.

Sample Output Table

Table 1 shows sample rows from the generated results table. Only selected columns are shown for readability.

Table 1: Sample diagnostic rows from File 05F SGA attack experiment.

Round	Attack	Malicious Client	FL Attack	SFL Attack	α_{FL}	α_{SFL}	FL Acc.	SFL Acc.
1	True	0	full_model_sga	client_side_gradient_reversal	0.01	20.0	0.1746	0.0980
2	True	0	full_model_sga	client_side_gradient_reversal	0.01	20.0	0.1902	0.0980
3	True	0	full_model_sga	client_side_gradient_reversal	0.01	20.0	0.1534	0.0980
96	True	0	full_model_sga	client_side_gradient_reversal	0.01	20.0	~ 0.10	0.3739
97	True	0	full_model_sga	client_side_gradient_reversal	0.01	20.0	~ 0.10	0.6250
100	True	0	full_model_sga	client_side_gradient_reversal	0.01	20.0	~ 0.10	0.3414

Results and Interpretation

Figure 24 shows that FL collapses rapidly under the SGA attack. The FL accuracy falls to approximately 0.10, which is close to random guessing for a 10-class classification problem. This confirms that full-model poisoning in FL is highly destructive because the attacker controls all layers.

In contrast, SFL does not fully collapse. The SFL accuracy is highly unstable and oscillatory, but it continues to recover during later rounds and reaches values above 0.60 in some rounds. This shows that SFL has a natural partial robustness because the attacker only controls the client-side layer.

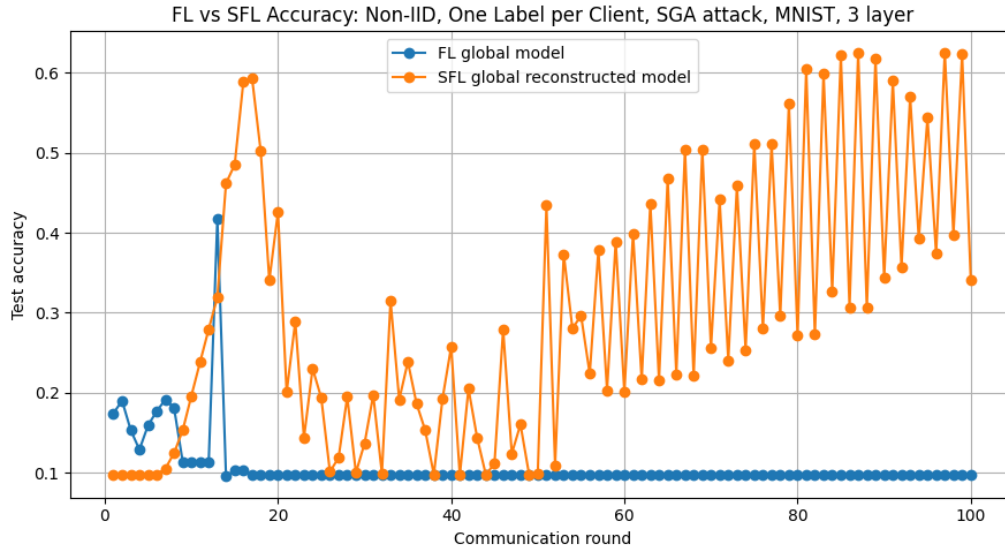


Figure 24: FL vs reconstructed SFL global test accuracy under the SGA poisoning attack. FL collapses toward random accuracy, while SFL remains unstable but continues partial learning.

Figure 25 confirms the same behaviour from the loss perspective. FL loss explodes rapidly, indicating numerical and optimisation collapse. SFL loss increases and becomes unstable around the middle rounds, but it later returns to a more controlled range. This supports the conclusion that the SFL attack is weaker because it is restricted to the client-side layer.

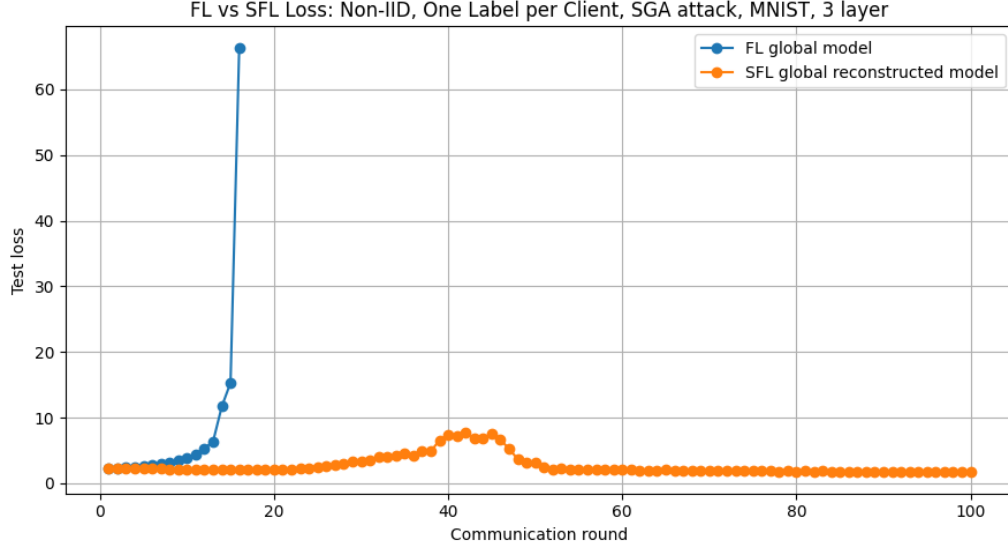


Figure 25: FL vs reconstructed SFL global test loss under the SGA poisoning attack. FL loss diverges catastrophically, while SFL avoids complete collapse despite instability.

Figure 26

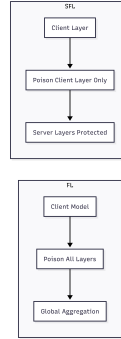


Figure 26: Comparison of SGA poisoning in FL and SFL architectures.

Overall, File 05F demonstrates that FL and SFL behave very differently under poisoning attacks. In the clean experiments, FL and SFL remained mathematically equivalent. In this attack experiment, the systems naturally diverge because the attack is applied to different parts of the model. This result motivates the later defence experiments, where robust aggregation and SFL-aware defence mechanisms are introduced.

File 05G: Median/MAD Defence against SGA Poisoning in FL and SFL

File 05G introduces the first defence mechanism against the poisoning attacks introduced in File 05F. The objective of this experiment was to investigate whether a lightweight robust aggregation defence could stabilise both Federated Learning (FL) and Split Federated Learning (SFL) under the same Scale Gradient Attack (SGA) conditions.

The experiment continues using:

- the MNIST dataset,
- the extreme one-label-per-client non-IID configuration,
- the three-layer MLP architecture,
- and the same SGA poisoning attack from File 05F.

The model architecture remains:

$$784 \rightarrow 256 \rightarrow 128 \rightarrow 10$$

with the SFL split defined as:

$$\text{Client side: } 784 \rightarrow 256$$

$$\text{Server side: } 256 \rightarrow 128 \rightarrow 10$$

The major change introduced in this notebook is the addition of a Median/MAD-based robust defence mechanism.

Why Median was Selected Instead of Mean

Traditional Federated Averaging uses the arithmetic mean:

$$\theta^{t+1} = \frac{1}{K} \sum_{k=1}^K \theta_k^{t+1}$$

However, the arithmetic mean is highly sensitive to outliers. A single malicious client producing extremely large poisoned updates can dominate the aggregation process.

The median operator was selected because it is substantially more robust against outliers. Instead of averaging updates directly, the defence computes the coordinate-wise median across client updates:

$$m_j = \text{median}(u_{1,j}, u_{2,j}, \dots, u_{K,j})$$

where:

- $u_{k,j}$ is parameter coordinate j from client k .

The defence then measures how far each client deviates from the median update. This approach was selected because:

- the project uses only 10 clients,
- and many advanced FL defences such as Krum, Multi-Krum, or Bulyan perform poorly or become unstable under small client counts.

Many established FL defence algorithms assume:

- large numbers of clients,
- or large proportions of honest participants.

Under extreme non-IID distributions with only 10 clients:

- naturally different client updates already appear highly anomalous,
- making neighbour-based FL defences unreliable.

Median/MAD was therefore selected because it:

- is lightweight,
- robust to extreme values,
- computationally simple,
- and does not require pairwise client-distance calculations.

MAD-Based Anomaly Detection

The defence uses the Median Absolute Deviation (MAD) statistic to identify suspicious updates.

First, the coordinate-wise median update is computed:

$$m = \text{median}(u_1, u_2, \dots, u_K)$$

Then the deviation of each client from the median is calculated:

$$d_k = \|u_k - m\|$$

The MAD statistic is then:

$$MAD = \text{median}(|d_k - \text{median}(d)|)$$

Each client receives a robust anomaly score:

$$z_k = \frac{|d_k - \text{median}(d)|}{MAD + \epsilon}$$

A client is flagged if:

$$z_k > \tau$$

where:

- τ is the MAD threshold coefficient.

In this notebook:

$$\tau = 6$$

The value was selected experimentally through repeated trial-and-error testing. Smaller thresholds caused:

- honest clients to be removed excessively under extreme non-IID conditions.

Larger thresholds:

- failed to suppress the poisoning sufficiently.

The selected value therefore represented a practical balance between:

- robustness,
- and preservation of legitimate client diversity.

How Defence Works in FL

In FL, the malicious client poisons all three layers of the model.

Therefore, the FL defence evaluates:

- the full model update vector,
- across all layers simultaneously.

The malicious client update is completely excluded from aggregation if flagged:

$$\mathcal{K}_{keep} = \{k \mid z_k \leq \tau\}$$

The robust FL aggregation becomes:

$$\theta^{t+1} = \frac{1}{|\mathcal{K}_{keep}|} \sum_{k \in \mathcal{K}_{keep}} \theta_k^{t+1}$$

Thus, flagged FL clients contribute nothing to the next global model.

How Defence Works in SFL

The SFL defence behaves differently because the attack only affects:

- the client-side layer.

The server-side layers are not directly poisoned by the malicious client. The SFL defence therefore evaluates:

- server-observable gradients,

- client-side parameter updates,
- and server-side optimisation behaviour.

Importantly:

- the SFL server never directly accesses client raw data,
- preserving privacy.

Instead, the server analyses:

- activations,
- gradients,
- and update statistics.

This allows anomaly detection without reconstructing private client datasets. The SFL client-side FL aggregation still occurs separately:

$$\theta_c^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_{c,k}^{t+1}$$

while server-side parameters are aggregated independently:

$$\theta_s^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_{s,k}^{t+1}$$

The reconstructed SFL model is then formed from both aggregated components.

Flagged Client Diagnostics

The notebook records:

FL flagged clients

and

SFL flagged clients

These diagnostics indicate which clients were classified as anomalous during each communication round.

This became extremely useful during debugging and hyperparameter tuning because:

- false positives could be identified,
- aggressive thresholds could be detected,
- and defence instability could be analysed across rounds.

The logs also allowed identification of:

- rounds where the malicious client escaped detection,
- or rounds where too many honest clients were removed.

Maximum Absolute Weight Monitoring

The experiment also records:

SFL max abs weight

and corresponding FL maximum absolute weights.

This metric monitors:

$$\max_i |\theta_i|$$

across all model parameters.

This diagnostic was important because poisoning attacks often create:

- exploding parameter values,
- numerical instability,
- NaNs,
- or optimisation divergence.

Monitoring maximum absolute weights helped identify:

- early instability,
- exploding gradients,
- and ineffective defence thresholds.

FL Minus SFL Accuracy

The metric:

$$A_{diff} = A_{FL} - A_{SFL}$$

was retained to compare robustness between systems.

Negative values indicate:

- SFL outperforming FL.

Positive values indicate:

- FL outperforming SFL.

This became useful for analysing:

- whether the defence stabilised one system more effectively than the other.

Role of the Global L2 Distance

The global L2 distance was retained from earlier experiments:

$$L_2 = \|\theta_{FL} - \theta_{SFL}\|_2$$

Although FL and SFL are no longer expected to remain equivalent under attack conditions, the L2 metric still provides useful information:

- small L2 growth suggests similar optimisation trajectories,
- while large L2 divergence indicates fundamentally different optimisation behaviour.

Under the defence setting:

- the L2 distance no longer represents equivalence validation,
- but instead measures behavioural divergence between defended FL and defended SFL systems.

Results and Interpretation

Figure 27 shows a major improvement compared to the undefended attack experiment in File 05F.

The FL model no longer collapses catastrophically. Instead:

- both FL and SFL continue stable learning,
- and both systems reach accuracies above 0.60.

Interestingly:

- SFL often slightly outperforms FL during many communication rounds.

This suggests that:

- the isolated attack surface in SFL,
- combined with robust aggregation,

creates a naturally stabilised optimisation process.

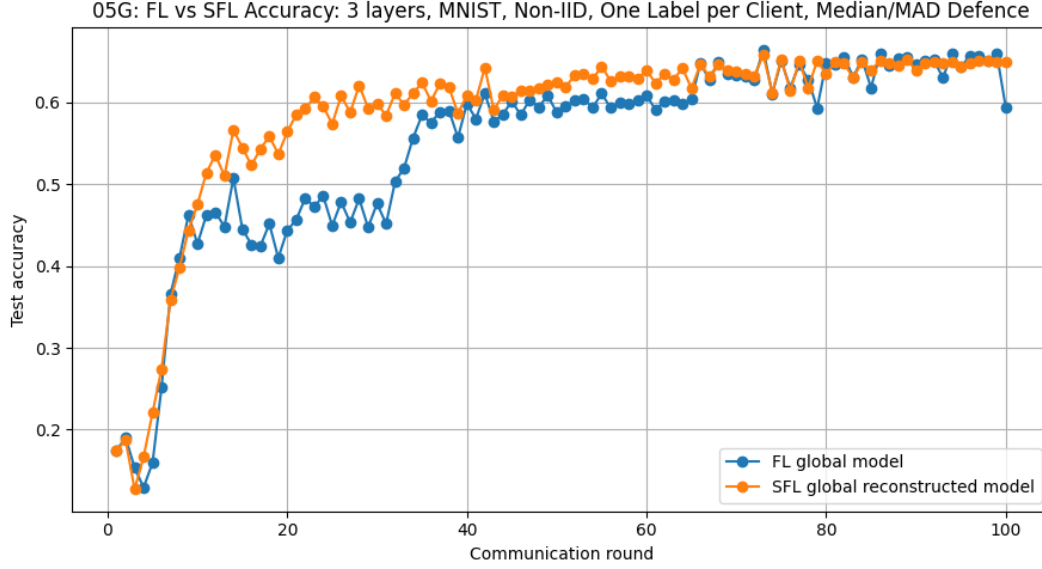


Figure 27: FL vs reconstructed SFL global test accuracy using the Median/MAD defence under SGA poisoning attacks. Both systems remain stable and avoid catastrophic collapse.

Figure 28 shows that:

- the FL loss remains unstable and oscillatory,
- while the SFL loss remains comparatively smoother.

This behaviour indicates that:

- although the defence successfully prevents collapse,
- FL still experiences stronger optimisation disruption because the attacker poisons the full model.

SFL remains more stable because:

- only the client-side layer is attacked,
- while the server-side optimisation remains comparatively protected.

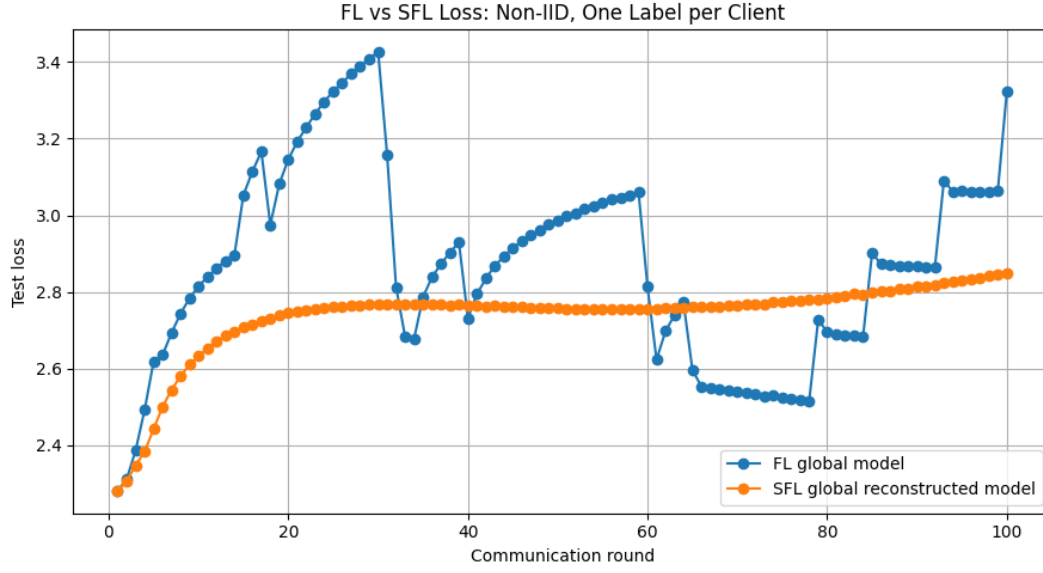


Figure 28: FL vs reconstructed SFL global test loss using the Median/MAD defence. FL remains noisier and less stable, while SFL maintains smoother optimisation behaviour.

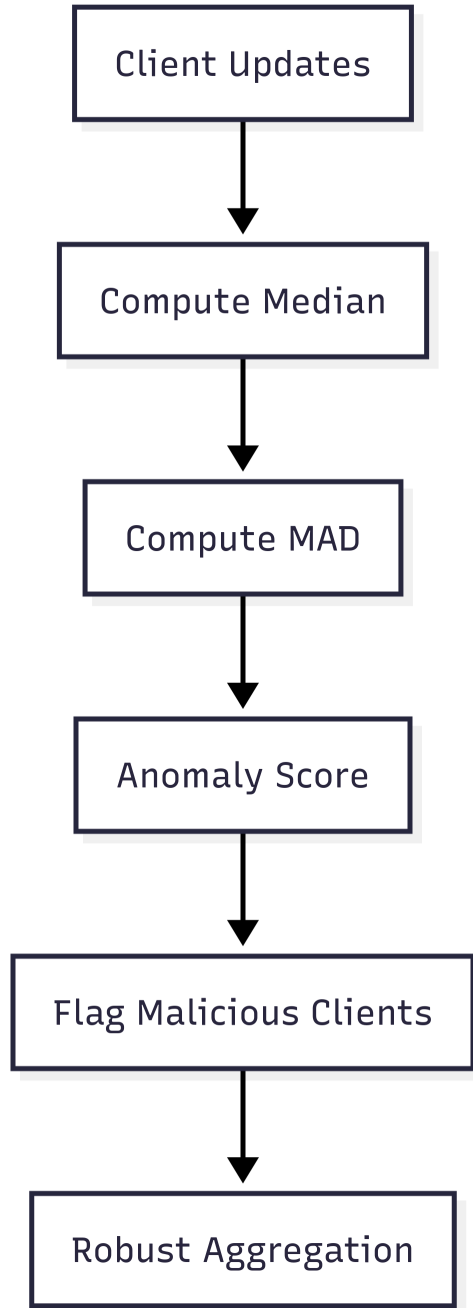


Figure 29: Median/MAD poisoning defence workflow.

Overall, File 05G demonstrates that:

- lightweight robust aggregation can significantly stabilise both FL and SFL under poisoning attacks,
- SFL naturally limits the attack surface,
- and Median/MAD defence mechanisms are particularly suitable for small-client extreme non-IID environments.

File 05H: Historical Cosine Similarity Defence with Warm-Up Phase

File 05H extends the poisoning defence investigation by replacing the Median/MAD defence from File 05G with a historical cosine-similarity-based anomaly detection mechanism. The objective of this experiment was to determine whether directional similarity between client updates across communication rounds could provide a more adaptive and privacy-preserving poisoning defence for both Federated Learning (FL) and Split Federated Learning (SFL).

The experiment continues using:

- the MNIST dataset,
- the extreme one-label-per-client non-IID configuration,
- the three-layer MLP architecture,
- and the same Scale Gradient Attack (SGA) poisoning attack introduced in File 05F.

The model architecture remains:

$$784 \rightarrow 256 \rightarrow 128 \rightarrow 10$$

with the SFL split defined as:

$$\text{Client side: } 784 \rightarrow 256$$

$$\text{Server side: } 256 \rightarrow 128 \rightarrow 10$$

Unlike File 05G, which used update magnitudes and Median/MAD robust statistics, File 05H attempts to detect poisoning based on directional inconsistency of client updates over time.

Motivation for Historical Cosine Defence

The motivation behind cosine-similarity-based defence was that poisoning attacks often change the direction of optimisation updates, not only their magnitude.

In normal training:

- client updates should gradually follow relatively consistent optimisation directions,
- especially when local data distributions remain stable across rounds.

A malicious client performing SGA poisoning introduces highly abnormal directional changes, even if update magnitudes are clipped or controlled.

Therefore, instead of measuring only update size, File 05H measures whether the direction of a client’s update remains consistent with its own historical behaviour.

Historical Warm-Up Phase

One major challenge in cosine-based defences is that early training rounds naturally contain unstable gradients.

During initial optimisation:

- honest client updates can differ significantly,
- particularly under extreme non-IID conditions.

To reduce false positives, a warm-up phase was introduced.

During the warm-up period:

- no clients are flagged,
- and historical memory vectors are gradually accumulated.

This allows the defence to first learn:

- the normal optimisation behaviour of each client.

Without this warm-up phase:

- the defence frequently classified honest clients as malicious during early rounds.

Mathematics of Cosine Similarity

For each client update vector:

$$u_k^{(t)}$$

the cosine similarity with the client's historical reference vector:

$$h_k^{(t-1)}$$

is computed as:

$$\cos(\theta_k) = \frac{u_k^{(t)} \cdot h_k^{(t-1)}}{\|u_k^{(t)}\| \|h_k^{(t-1)}\|}$$

where:

- values close to 1 indicate highly aligned updates,
- values near 0 indicate orthogonal behaviour,
- and negative values indicate opposite optimisation directions.

Clients with unusually low cosine similarity are considered suspicious.

Historical EMA Reference Vector

The historical reference vector is updated using an Exponential Moving Average (EMA):

$$h_k^{(t)} = \alpha h_k^{(t-1)} + (1 - \alpha) u_k^{(t)}$$

where:

- $h_k^{(t)}$ is the historical memory vector,
- $u_k^{(t)}$ is the current update,
- and α is the EMA smoothing coefficient.

This allows the defence to:

- retain long-term optimisation behaviour,
- while still adapting gradually to changing gradients.

The historical EMA mechanism was substantially more stable than:

- simple round-to-round cosine comparison,
- or direct global-reference cosine comparison.

Other Cosine Defences Attempted

Several alternative cosine-based approaches were tested before settling on the historical warm-up implementation:

1. Direct round-to-round cosine similarity

- compared each update only against the previous round,
- but became extremely unstable under non-IID distributions.

2. Global-model cosine comparison

- compared client updates against the global model update,
- but honest non-IID clients naturally diverged strongly from the global direction.

3. Pure cosine thresholding without warm-up

- produced excessive false positives during early rounds.

4. Fixed cosine thresholds

- failed because cosine distributions changed substantially throughout training.

The historical EMA warm-up version performed better because:

- it personalised the reference behaviour for each client,
- and reduced early-round instability.

How Cosine Defence was Applied in FL

In FL:

- the attack affects all three model layers,
- therefore the cosine similarity is computed on the full flattened model update vector.

For each FL client:

$$u_k^{FL} = \theta_k^{(t)} - \theta^{(t-1)}$$

The cosine score is computed using the client’s full update vector against its EMA historical memory.

Flagged clients are excluded from aggregation:

$$\mathcal{K}_{keep} = \{k \mid \cos(\theta_k) \geq \tau\}$$

The remaining honest clients are aggregated normally.

How Cosine Defence was Applied in SFL

In SFL:

- only the client-side layer is directly poisoned by the attacker.

Therefore:

- cosine similarity is computed only on the client-side FL update portion.

This means the defence evaluates:

- client-observable optimisation behaviour,
- without exposing raw data,
- activations,
- or labels.

Importantly:

- the SFL server still does not directly access private client datasets.

Only optimisation statistics are exchanged.

The SFL server-side aggregation remains separate:

$$\theta_s^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_{s,k}^{t+1}$$

while the client-side FL aggregation uses cosine-filtered updates.

Results and Interpretation

Compared to File 05F (no defence):

- File 05H clearly improves SFL robustness.

The SFL model:

- eventually recovers,
- and reaches approximately 0.58 – 0.60 accuracy.

This represents a major improvement over:

- the undefended poisoning experiment,
- where optimisation became substantially more unstable.

However, the FL system still collapses catastrophically.

The FL accuracy falls toward:

0.10

which corresponds approximately to random guessing on MNIST.

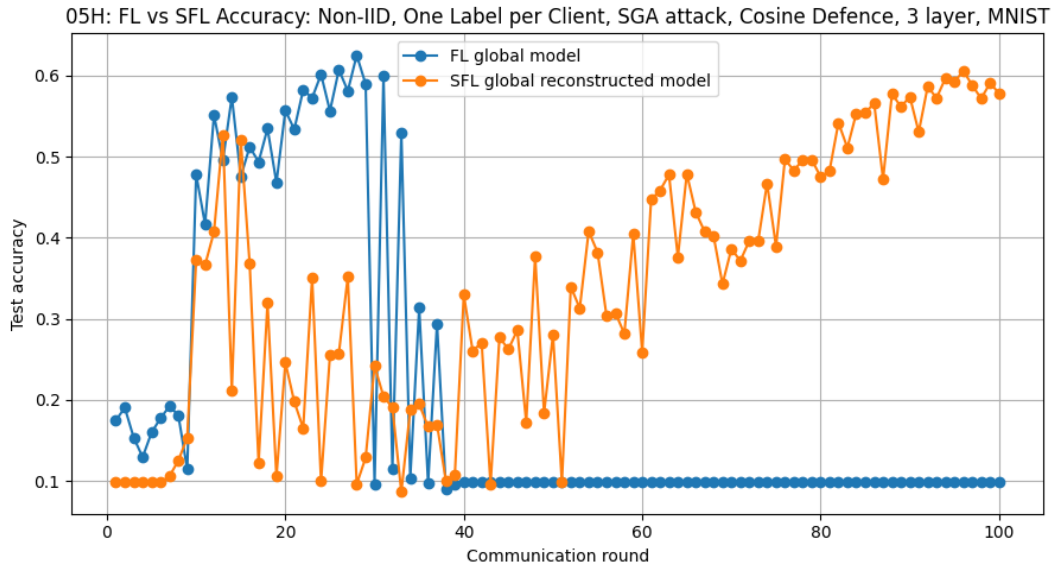


Figure 30: FL vs reconstructed SFL global test accuracy using the historical cosine defence with warm-up. SFL partially recovers, while FL still collapses under poisoning.

The FL loss also explodes dramatically:

loss $\rightarrow$ 70

indicating:

- severe optimisation divergence,
- and failure of the cosine defence to sufficiently isolate poisoned FL updates.

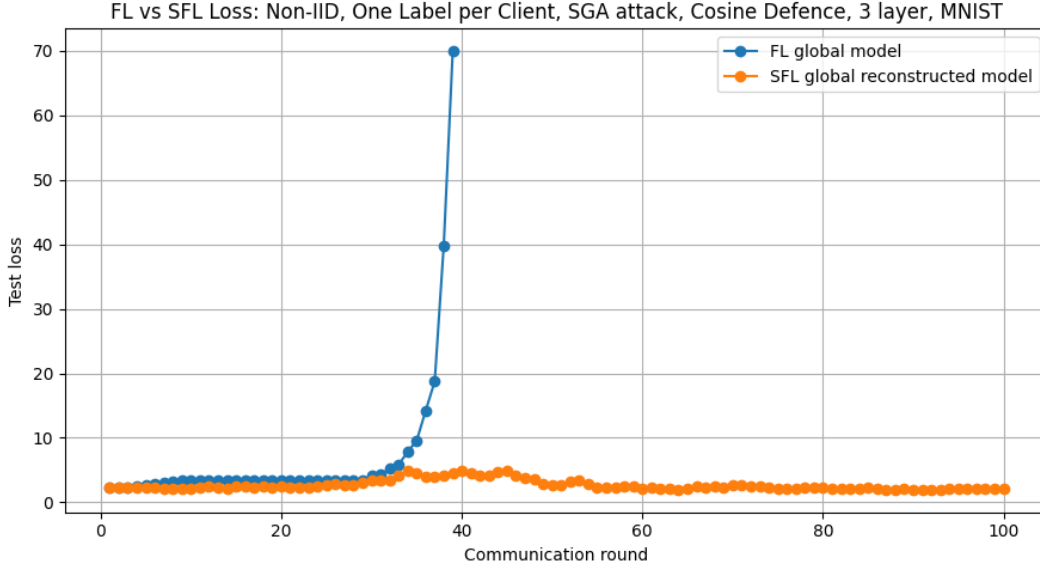


Figure 31: FL vs reconstructed SFL global test loss under historical cosine defence. FL diverges catastrophically while SFL remains comparatively stable.

Why Cosine Defence Still Did Not Work Well

Although cosine similarity improved SFL stability somewhat, the overall defence remained substantially weaker than the Median/MAD defence from File 05G.

Several reasons explain this:

1. **Extreme non-IID behaviour naturally changes update directions**
 - honest clients frequently appear directionally inconsistent.
2. **Cosine similarity ignores update magnitude**
 - a poisoned update can still align directionally while having destructive magnitude.
3. **Small client count**
 - only 10 clients were available,
 - reducing statistical robustness.
4. **Deep model splitting**
 - SFL client-side updates already contain reduced information compared to full FL updates.

5. Historical drift

- the EMA memory gradually adapts,
- meaning malicious updates can slowly contaminate historical references.

Overall, File 05H demonstrated that:

- cosine-based directional defences alone are insufficient under strong poisoning attacks and extreme non-IID settings,
- although SFL naturally remains more robust than FL,
- and historical warm-up mechanisms significantly outperform naive cosine-threshold methods.

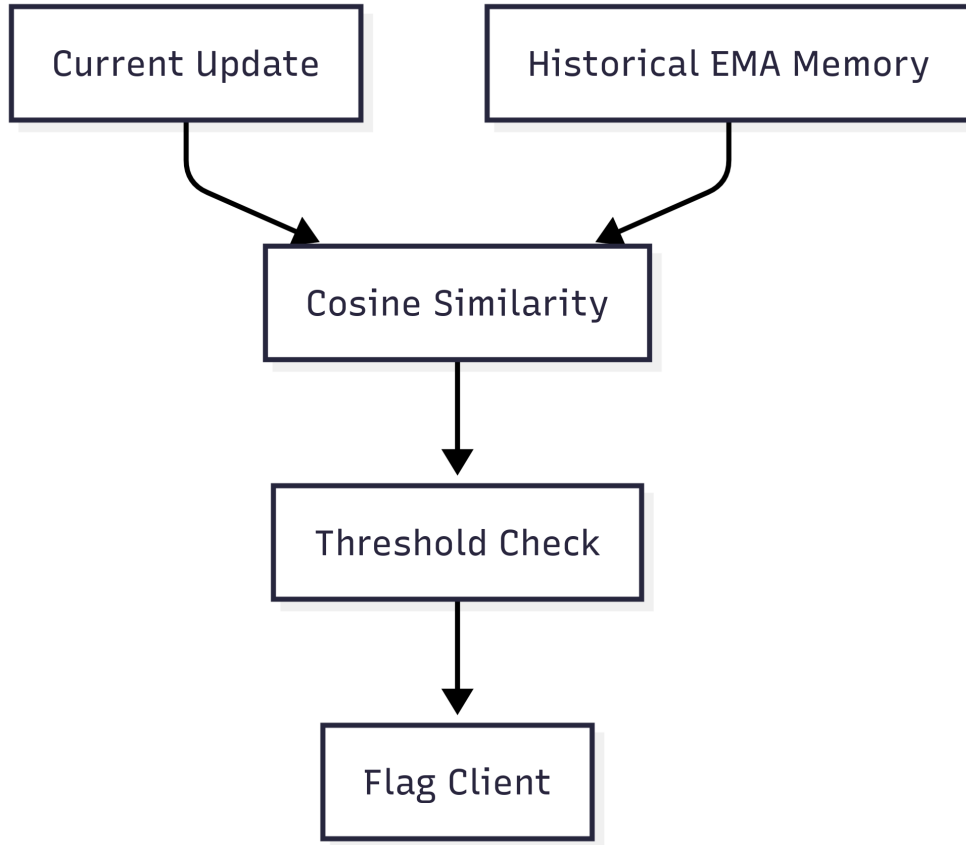


Figure 32: Historical cosine similarity defence using EMA memory vectors.

The experiment also provided strong motivation for the later combined defence approaches implemented in Files 05I and 05J, where:

- cosine similarity was integrated with Median/MAD statistics,
- rather than being used independently.

File 05I: Combined Median/MAD and Historical Cosine Defence

File 05I represents the first true hybrid defence model developed in this project. After independently evaluating:

- the Median/MAD statistical defence in File 05G,
- and the historical cosine-similarity defence in File 05H,

the next logical progression was to combine both approaches into a single defence framework.

The motivation behind this experiment was that:

- Median/MAD is effective at detecting abnormal update magnitudes,
- while cosine similarity is effective at detecting abnormal optimisation directions.

Therefore, combining both mechanisms was expected to provide:

- stronger robustness,
- lower false positives,
- and improved poisoning isolation under extreme non-IID conditions.

This file continues using:

- the MNIST dataset,
- the one-label-per-client extreme non-IID distribution,
- the three-layer MLP architecture,
- and the same SGA poisoning attack from Files 05F–05H.

The model structure remains:

$$784 \rightarrow 256 \rightarrow 128 \rightarrow 10$$

with the SFL split:

$$\text{Client side: } 784 \rightarrow 256$$

$$\text{Server side: } 256 \rightarrow 128 \rightarrow 10$$

Why a Combined Defence was Needed

The results from earlier experiments showed complementary strengths and weaknesses:

- File 05G (Median/MAD) produced strong stability and robust convergence,
- but sometimes failed to detect directionally abnormal updates.

Meanwhile:

- File 05H (historical cosine defence) could identify directional anomalies,
- but was unstable under extreme non-IID distributions,
- and still allowed FL collapse.

Therefore, File 05I combines:

1. magnitude-based anomaly detection,
2. and directional anomaly detection.

The intuition was:

A poisoned update should usually appear suspicious either in magnitude, direction, or both.

Median/MAD Component

As introduced in File 05G, the Median/MAD defence computes robust statistics over client update norms.

For each client update:

$$u_k$$

the update magnitude is computed:

$$s_k = \|u_k\|_2$$

The median score:

$$m = \text{median}(s_1, s_2, \dots, s_K)$$

and the Median Absolute Deviation:

$$MAD = \text{median}(|s_k - m|)$$

are then calculated.

Clients exceeding the threshold:

$$s_k > m + \lambda \cdot MAD$$

are considered suspicious.

This defence remains highly robust because:

- the median is insensitive to extreme poisoned updates,
- unlike mean-based aggregation.

Historical Cosine Component

The cosine component from File 05H is also retained.

Each update vector:

$$u_k^{(t)}$$

is compared against the client’s historical EMA reference vector:

$$h_k^{(t-1)}$$

using cosine similarity:

$$\cos(\theta_k) = \frac{u_k^{(t)} \cdot h_k^{(t-1)}}{\|u_k^{(t)}\| \|h_k^{(t-1)}\|}$$

The EMA historical memory is updated using:

$$h_k^{(t)} = \alpha h_k^{(t-1)} + (1 - \alpha) u_k^{(t)}$$

This allows the defence to:

- retain long-term optimisation behaviour,
- while still adapting gradually to changing gradients.

How Both Defences Were Combined

Importantly, File 05I is not simply a direct copy-and-paste combination of Files 05G and 05H.

Several implementation changes were required to stabilise the hybrid defence.

The combined logic works as follows:

1. Compute Median/MAD anomaly scores.
2. Compute historical cosine similarity scores.
3. Independently flag suspicious clients from each defence.
4. Merge the flagged-client sets.

The final flagged set becomes:

$$\mathcal{F} = \mathcal{F}_{MAD} \cup \mathcal{F}_{cosine}$$

Only the remaining clients are aggregated:

$$\mathcal{K}_{keep} = \{1, \dots, K\} \setminus \mathcal{F}$$

This means:

- a client can be removed if either defence detects suspicious behaviour.

This hybrid filtering substantially improves robustness.

How the Defence was Applied Differently in FL and SFL

The attack affects FL and SFL differently, therefore the defence also operates differently.

FL Defence In FL:

- the SGA attack modifies all model layers simultaneously.

Therefore:

- Median/MAD is computed on the full flattened model update,
- cosine similarity is also computed on the full update vector.

The FL global aggregation becomes:

$$\theta^{t+1} = \sum_{k \in \mathcal{K}_{keep}} \frac{n_k}{n} \theta_k^{t+1}$$

where poisoned clients are excluded.

SFL Defence In SFL:

- the attack only directly affects the client-side layer.

Therefore:

- the FL-style defence operates only on the client-side update,
- while server-side aggregation remains separate.

This preserves SFL privacy because:

- raw data never leaves the client,
- only client-side optimisation updates are analysed.

The server still aggregates server-side parameters independently:

$$\theta_s^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_{s,k}^{t+1}$$

Thus:

- the SFL defence remains split-aware,
- unlike standard FL defences.

Why This File is Different from 05G and 05H

Although File 05I combines concepts from both earlier files, it is not merely their direct union.

The major differences include:

1. joint decision logic between MAD and cosine signals,
2. unified client filtering,
3. stabilised EMA updates,
4. improved warm-up handling,
5. and revised threshold behaviour.

Additionally:

- the cosine defence no longer acts independently,
- but instead acts as a secondary verification signal.

This substantially reduces:

- false positives,
- and instability seen in File 05H.

Results and Interpretation

The combined defence performs substantially better than the standalone cosine defence.

Unlike File 05H:

- the FL system no longer catastrophically collapses,
- and both FL and SFL remain stable throughout training.

The final accuracies converge near:

$$0.62 - 0.65$$

for both systems.

05I: FL vs SFL Accuracy: Non-IID, One Label per Client, Defence Enabled (Combined Median/MAD + Historical Cosine)

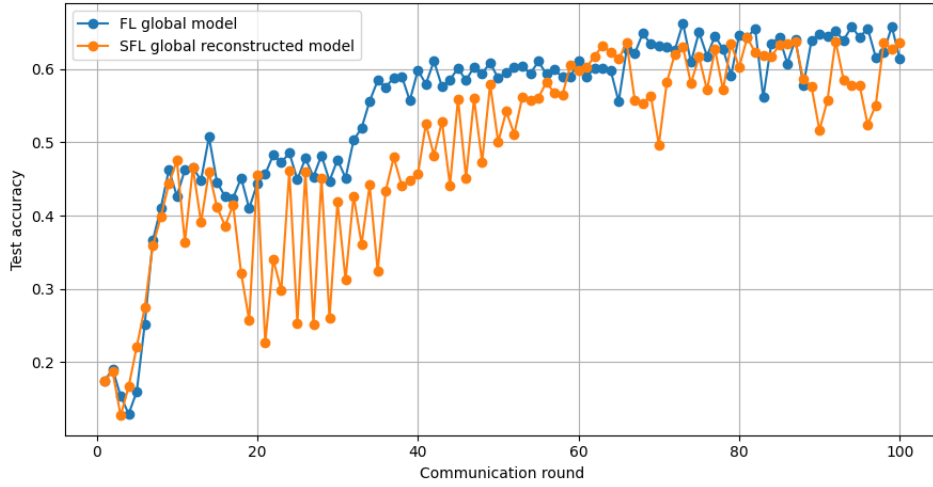


Figure 33: FL and reconstructed SFL accuracy under the combined Median/MAD and historical cosine defence. Both systems remain stable and converge successfully.

Compared to:

- File 05F (no defence),
- and File 05H (cosine-only defence),

the hybrid defence:

- produces dramatically improved robustness,
- significantly reduces FL collapse,
- and maintains strong SFL convergence.

The loss curves also remain substantially more stable:

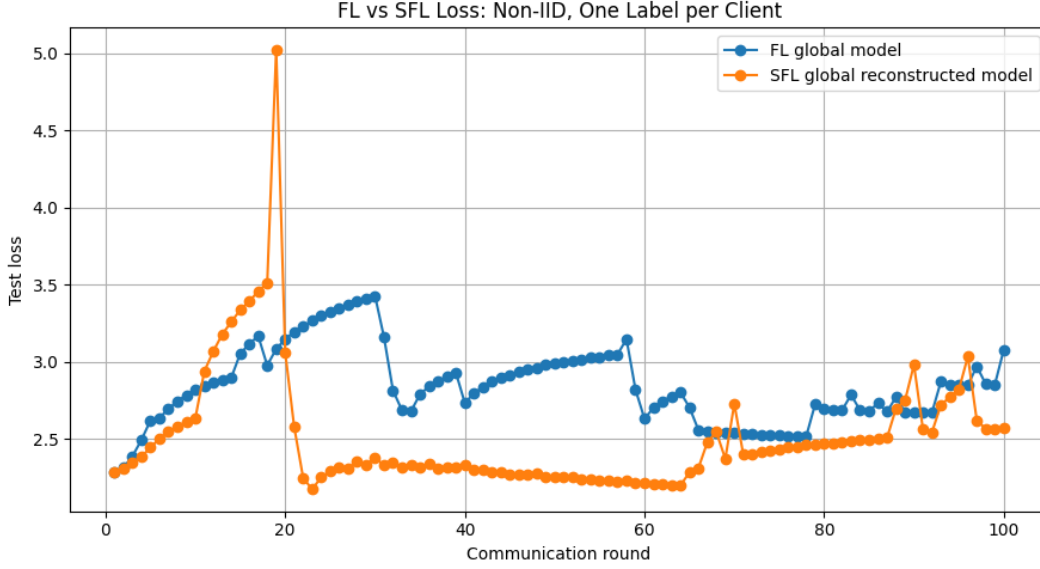


Figure 34: FL and reconstructed SFL loss under the combined defence. Unlike previous cosine-only experiments, both systems remain numerically stable.

Unlike File 05H:

- no catastrophic FL divergence occurs,
- and the combined defence successfully prevents exploding optimisation.

Accuracy Graph Analysis and Comparison with Previous Defences

The accuracy curves in File 05I demonstrate that the combined defence produces substantially more stable optimisation behaviour than the standalone cosine defence from File 05H, while also maintaining comparable robustness to the Median/MAD defence from File 05G.

Compared to File 05H:

- the FL model no longer collapses toward random guessing,
- the catastrophic divergence previously observed in FL is eliminated,
- and both FL and SFL remain within a stable convergence region throughout training.

In File 05H, the cosine-only defence was unable to sufficiently isolate poisoned FL updates. As a result:

- FL accuracy collapsed toward approximately 0.10,
- while FL loss exploded toward values above 70.

In contrast, File 05I maintains:

- FL accuracy near 0.62 – 0.65,

- and SFL accuracy near $0.62 - 0.64$,

indicating that the combined defence successfully stabilised both systems.

The SFL accuracy curve in File 05I also shows reduced oscillation compared to File 05H. Although some fluctuations remain due to:

- the extreme non-IID setting,
- and the aggressive poisoning attack,

the optimisation trajectory is substantially smoother than the cosine-only experiment. Compared to File 05G (Median/MAD only):

- the final accuracies are relatively similar,
- however the combined defence provides additional directional anomaly filtering,
- which improves robustness against poisoned updates that may have normal magnitude but abnormal optimisation direction.

The combined defence therefore acts as:

- a stronger general-purpose defence than cosine-only methods,
- while also providing more comprehensive anomaly detection than Median/MAD alone.

Overall, the graphs demonstrate that:

1. Median/MAD provides strong statistical stability,
2. cosine similarity provides directional anomaly awareness,
3. and combining both mechanisms produces the most balanced behaviour observed so far in the project.

Key Conclusion

File 05I demonstrated that:

- combining robust statistical filtering with directional historical analysis is substantially more effective than using either mechanism independently.

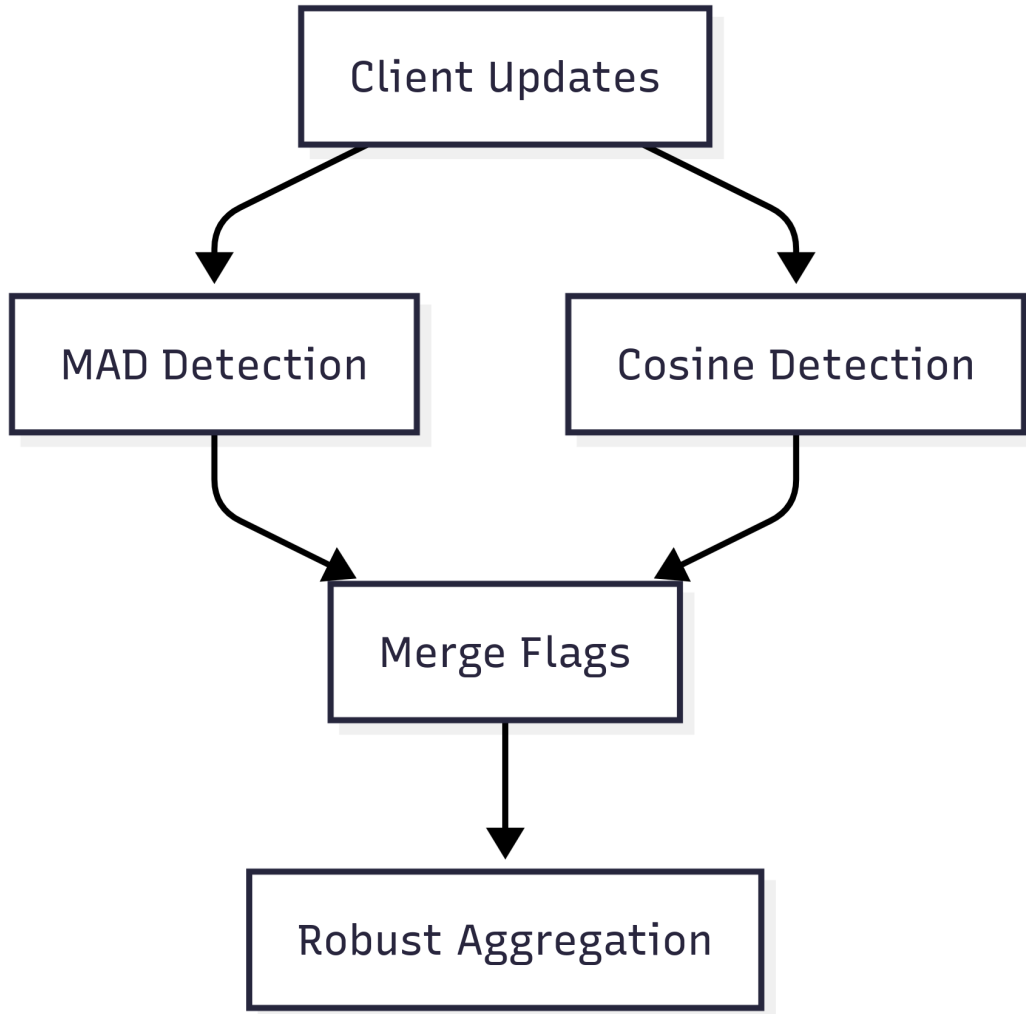


Figure 35: Combined Median/MAD and historical cosine defence workflow.

The experiment also showed that:

- split-aware poisoning defence for SFL requires fundamentally different treatment from traditional FL,
- because the attack surface itself is structurally different.

Most importantly:

- the hybrid defence preserved SFL privacy constraints,
- while still providing strong poisoning robustness under extreme non-IID conditions.

File 05J: SFL-Aware Median/MAD Defence using Server-Observable Loss

File 05J investigates a more architecture-aware defence mechanism for Split Federated Learning (SFL). After File 05I showed that combining Median/MAD and historical cosine could improve robustness compared with standalone cosine, File 05J moves in a different direction. Instead of adding more anomaly signals, this file focuses on where the defence should be applied in FL and SFL, given that the poisoning attack affects the two systems differently.

The experiment continues using:

- the MNIST dataset,
- the extreme one-label-per-client non-IID setting,
- the three-layer MLP architecture,
- and the same SGA poisoning attack on client 0.

The model remains:

$784 \rightarrow 256 \rightarrow 128 \rightarrow 10$

with the SFL split:

Client side: $784 \rightarrow 256$

Server side: $256 \rightarrow 128 \rightarrow 10$

Important Clarification: Does File 05J Use Cosine?

Although the notebook still contains a cosine helper function, the active defence logic in the main experiment does not use cosine for the SFL filtering decision. In the main training loop, the SFL defence is based on server-observable loss scores:

```
sfl_defence_mode = median_loss_observable
```

and the recorded cosine scores are set to:

```
sfl_cosine_scores = None
```

Therefore, File 05J should not be described as a cosine-combined defence. It is better described as:

an SFL-aware Median/MAD defence based on server-observable loss, combined with full-model Median/MAD update filtering for FL.

This is different from File 05I, which genuinely combines Median/MAD and historical cosine similarity.

Motivation for File 05J

The main motivation behind File 05J is that FL and SFL have different attack surfaces.

In standard FL, the malicious client controls the full model update:

$$\Delta\theta_{FL}$$

which includes all layers of the network.

In SFL, the malicious client only controls the client-side layer:

$$\Delta\theta_c$$

while the server-side model parameters:

$$\theta_s$$

are not directly controlled by the client.

This means that applying the same defence logic to both FL and SFL may not be optimal. File 05J therefore implements defence logic that is adapted to the structure of each system.

FL Defence in File 05J

For FL, the defence is applied to the full model update because the FL attacker can poison all layers.

For each FL client, the update distance from the pre-round global model is calculated:

$$s_k^{FL} = \|\theta_k^{t+1} - \theta^t\|_2$$

where:

- θ_k^{t+1} is the locally trained model of client k ,
- and θ^t is the global model before the round.

Median/MAD filtering is then applied to the update scores:

$$m^{FL} = \text{median}(s_1^{FL}, s_2^{FL}, \dots, s_K^{FL})$$

$$MAD^{FL} = \text{median}(|s_k^{FL} - m^{FL}|)$$

A client is flagged if:

$$s_k^{FL} > m^{FL} + \lambda MAD^{FL}$$

where λ is the MAD multiplier.

The FL aggregation then keeps only the non-flagged clients:

$$\theta^{t+1} = \sum_{k \in \mathcal{K}_{keep}^{FL}} \frac{n_k}{n} \theta_k^{t+1}$$

This is a full-model robust aggregation defence.

SFL Defence in File 05J

The SFL defence is different because the server does not directly inspect client-side weights. Instead, it uses server-observable behaviour during split learning.

In this file, the active SFL defence uses the per-client server-side loss:

$$s_k^{SFL} = \bar{\ell}_k$$

where $\bar{\ell}_k$ is the mean server-observed loss for client k during that communication round. Median/MAD filtering is then applied to these loss scores:

$$m^{SFL} = \text{median}(s_1^{SFL}, s_2^{SFL}, \dots, s_K^{SFL})$$

$$MAD^{SFL} = \text{median}(|s_k^{SFL} - m^{SFL}|)$$

A client is flagged if:

$$s_k^{SFL} > m^{SFL} + \lambda MAD^{SFL}$$

The flagged SFL clients are then excluded from the client-side aggregation:

$$\theta_c^{t+1} = \sum_{k \in \mathcal{K}_{keep}^{SFL}} \frac{n_k}{n} \theta_{c,k}^{t+1}$$

However, the server-side aggregation is intentionally handled differently. In File 05J, the server-side model aggregation keeps all server-side states:

$$\theta_s^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_{s,k}^{t+1}$$

This is one of the key split-aware design decisions in the file.

Key Difference Between 05I and 05J

File 05I and File 05J are not the same type of combined defence.

File 05I combines multiple anomaly signals:

$$\mathcal{F}_{05I} = \mathcal{F}_{MAD} \cup \mathcal{F}_{cosine}$$

where:

- $\mathcal{F}_{MAD}$ is the set of clients flagged by Median/MAD,
- and $\mathcal{F}_{cosine}$ is the set of clients flagged by cosine similarity.

Therefore, File 05I is a multi-signal defence.

File 05J is different. It does not actively combine cosine with MAD in the main decision logic. Instead, it changes the defence location and granularity:

FL: full-model Median/MAD

SFL: server-observable loss Median/MAD

The two key differences between 05I and 05J are therefore:

1. 05I combines detection signals, while 05J changes the defended region.

File 05I asks whether combining Median/MAD with cosine improves detection. File 05J asks whether SFL should be defended differently because the attack surface is split across client-side and server-side components.

2. 05J uses server-observable SFL loss rather than historical cosine.

In 05J, the SFL server does not need to inspect client weights or use cosine history. It only uses loss values already visible during server-side split learning.

How Privacy is Preserved

The SFL server does not access:

- raw client data,
- full client-side model weights,
- or private labels beyond the normal training loss calculation.

The defence uses server-observable signals generated during SFL training. In particular, the server observes:

- smashed activations,
- server-side losses,
- and gradients required for split backpropagation.

Thus, the defence does not require converting SFL back into full FL or exposing the full client model.

Accuracy Graph Analysis

The accuracy graph shows that File 05J produces one of the strongest overall results observed so far.

05J: FL vs SFL Accuracy: Non-IID, One Label per Client, attack with SGA attack on client 0, defence with combined median/MAD on server-observable loss and gradients

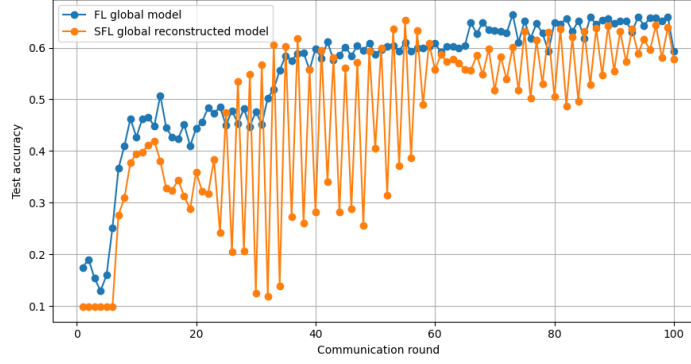


Figure 36: FL and reconstructed SFL accuracy under the SFL-aware Median/MAD defence using server-observable loss.

The FL model remains stable and converges to approximately:

$$0.64 - 0.66$$

This is a major improvement compared with:

- File 05F, where FL collapsed without defence,
- and File 05H, where FL collapsed under cosine-only defence.

The SFL model also improves and reaches approximately:

$$0.58 - 0.64$$

although its accuracy remains more oscillatory than FL. The oscillation is expected because:

- the data distribution is extremely non-IID,
- only one label exists per client,
- and the SFL defence filters the client-side aggregation based on server-observable loss.

Compared with File 05G, File 05J achieves broadly comparable final accuracy, but its logic is more SFL-specific. Compared with File 05I, the FL curve remains similarly strong, while the SFL curve is still oscillatory but benefits from an architecture-aware filtering design.

Loss Graph Analysis

The loss graph shows a particularly important difference between FL and SFL.

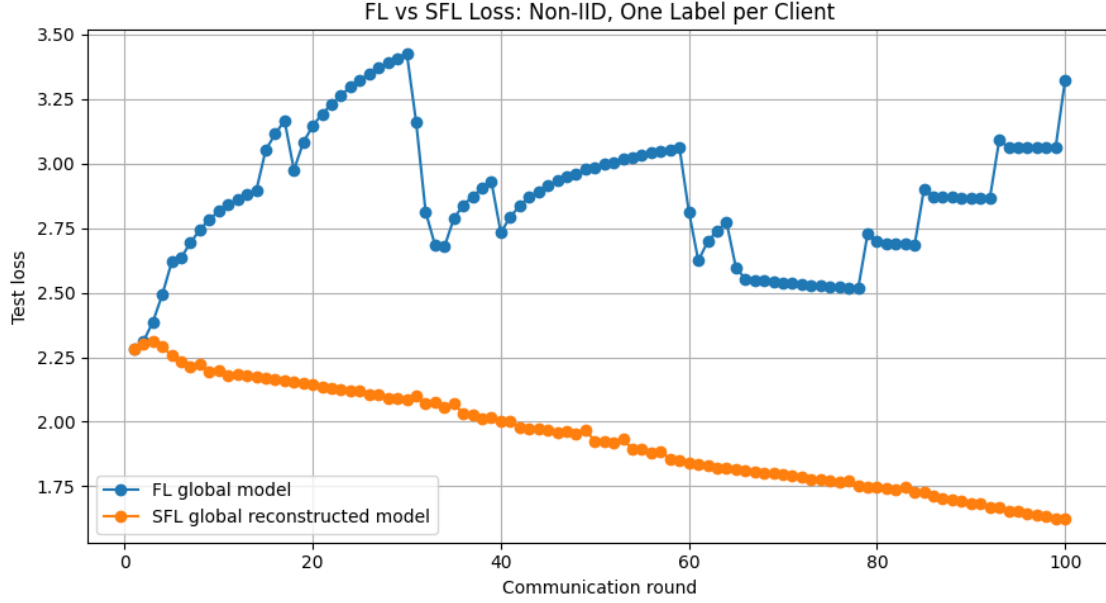


Figure 37: FL and reconstructed SFL loss under the SFL-aware Median/MAD defence. SFL loss decreases steadily, while FL loss remains more oscillatory.

The SFL loss decreases steadily across communication rounds. This suggests that even though the SFL accuracy oscillates, the model is gradually improving its confidence and optimisation stability.

The FL loss remains higher and more variable. This is logical because the FL attack affects the full model update, so even with Median/MAD filtering, the full-model optimisation trajectory remains more exposed to poisoned behaviour.

Comparison with Previous Defences

Compared with File 05F:

- 05J clearly prevents full collapse.

Compared with File 05H:

- 05J performs much better for FL,
- because it does not rely on unstable cosine direction assumptions.

Compared with File 05G:

- 05J produces similar final robustness,
- but introduces a more SFL-aware defence structure.

Compared with File 05I:

- 05I is a true multi-signal defence,

- while 05J is a split-aware server-observable defence.

The results are logical because cosine similarity becomes unreliable under one-label-per-client non-IID data. Honest clients naturally produce different update directions. By contrast, server-observable loss is a more directly meaningful signal in SFL because a malicious client should tend to create abnormal server-side training behaviour.

Key Conclusion

File 05J should be interpreted as an SFL-aware defence experiment rather than a cosine-combined defence.



Figure 38: SFL-aware defence using server-observable loss statistics.

Its main contribution is that it demonstrates:

SFL defence does not need to copy FL defence directly. Instead, it can exploit server-observable signals and the split architecture to identify suspicious clients while preserving privacy.

The two strongest findings are:

1. Full-model Median/MAD remains effective for FL because the FL attacker controls the full model.
2. Server-observable Median/MAD loss filtering is meaningful for SFL because the server can detect abnormal behaviour without inspecting client-side weights.

This makes File 05J an important step toward architecture-aware poisoning defence for Split Federated Learning.

Overall Defence Comparison (Files 05G–05J)

The four defence experiments demonstrated important differences between statistical and directional anomaly detection methods under extreme non-IID Split Federated Learning conditions.

Among all tested approaches, the Median/MAD defence in File 05G produced the most stable and reliable overall behaviour. Both FL and SFL maintained stable convergence under the SGA poisoning attack, and the method remained robust despite the small number of clients and highly heterogeneous data distribution.

The historical cosine defence in File 05H showed the weakest performance. Although the cosine-based approach improved SFL robustness compared with the undefended attack scenario, it failed to prevent FL collapse. This occurred because extreme non-IID training naturally produces highly different update directions between honest clients, making cosine similarity less reliable as an anomaly signal.

File 05I combined Median/MAD and historical cosine similarity into a multi-signal defence framework. This improved performance compared with cosine-only defence and maintained strong FL robustness. However, the combined method did not clearly outperform the standalone Median/MAD defence, suggesting that cosine similarity contributed limited additional value under the one-label-per-client setting.

File 05J introduced a split-aware SFL defence using server-observable loss statistics rather than direct cosine filtering. Although its final accuracy remained similar to 05G, the experiment demonstrated an important conceptual contribution: SFL defence mechanisms can exploit server-visible behaviour without requiring access to the full client model or violating privacy assumptions.

Overall, the experiments suggest that robust statistical filtering methods such as Median/MAD are more reliable than directional cosine-based methods under extreme non-IID conditions with small client counts. Furthermore, the results indicate that SFL-specific defence mechanisms should consider the split architecture and observable server-side behaviour rather than directly reusing traditional FL defence strategies.